

CHAPTER 2 Unit 2

Overall Heat Transfer Coefficient and Insulation

Syllabus :

Thermal contact resistance, Critical radius of insulation, Electrical analogy, overall heat transfer coefficient, Heat Sources

Syllabus Topic : Variable Thermal Conductivity

2.1 Heat Conduction in Thick Wall with Variable Thermal Conductivity

The dependence of thermal conductivity on temperature can be expressed as :

$$k = k_0 (1 + \alpha \cdot T)$$

Where k_0 is the thermal conductivity at reference temperature, α is constant for a given material and T is the temperature.

Fouriers law of heat conduction through a thick is expressed as :

$$Q = -kA \cdot \frac{dT}{dx} = -k_0 (1 + \alpha \cdot T) A \frac{dT}{dx}$$

$$\therefore \frac{Q}{A} \cdot dx = k_0 (1 + \alpha \cdot T) \cdot dT \quad \dots(i)$$

Integrating the above equation with boundary conditions at $x = 0, T = T_1$ and at $x = x, T = T_2$.

$$\frac{Q}{A} \int_0^x dx = - \int_{T_1}^{T_2} k_0 (1 + \alpha T) \cdot dT$$

$$\frac{Q}{A} [x]_0^x = -k_0 \left[T + \frac{\alpha}{2} T^2 \right]_{T_1}^{T_2}$$

$$\frac{Q}{A} (x - 0) = -k_0 \left[(T_2 - T_1) + \frac{\alpha}{2} (T_2^2 - T_1^2) \right]$$

$$= k_0 \left[(T_1 - T_2) + \frac{\alpha}{2} (T_2 - T_1) (T_1 + T_2) \right]$$

$$\frac{Q}{A} \cdot x = +k_0 (T_1 - T_2) \left[1 + \frac{\alpha}{2} (T_1 + T_2) \right] \dots(2.1.1)$$

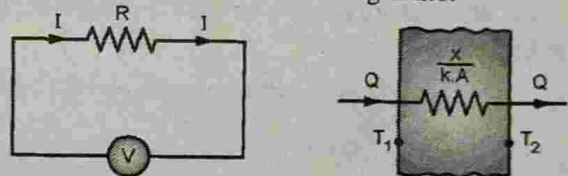
$$Q = \frac{k_m \cdot A (T_1 - T_2)}{x} \quad \dots(2.1.2)$$

Where $k_m = k_0 \left[1 + \frac{\alpha}{2} (T_1 + T_2) \right]$ represents the mean value of thermal conductivity calculated at mean temperature of $T = \left(\frac{T_1 + T_2}{2} \right)$

Electrical Analogy

2.2 Analogy between Heat Conduction and Electricity

Rate of heat flow has an analogy with current flow in an electrical system having the electrical resistance R with potential difference V as shown in Fig. 2.2.1.



(a) Heat conduction

(b) Electrical system

Fig. 2.2.1 : Analogy between electrical conduction and heat conduction system

From ohm's law for electrical system we can write,

$$\therefore \text{Current flow, } I = \frac{V}{R} \quad \dots(2.2.1)$$

For heat conduction system,

$$Q = \frac{k \cdot A (T_1 - T_2)}{x} = \frac{(T_1 - T_2)}{\left(\frac{x}{kA} \right)} = \frac{\Delta T}{R} \quad \dots(2.2.2)$$

By comparing Equations (2.2.1) and (2.2.2) we draw the following analogy between electricity and heat flow.

- Temperature difference $(T_1 - T_2)$ across the wall represents the driving force equivalent to potential difference, V .
- Heat flow rate Q corresponds to current flow I .

- (iii) $\left(\frac{x}{kA}\right)$ represents the **thermal resistance to heat flow rate**, R equivalent to electrical resistance R .

2.2.1 Procedure of Applying Electrical Analogy to Heat Transfer Problems

Following procedure is followed to determine the heat transfer rates in composite structures.

- Draw schematic diagram of the system.
- Draw an equivalent electric circuit of the system.
- Compute all the thermal resistances in the circuit.
- Calculate the equivalent resistance, ΣR .
- Using equation $Q = \frac{T_i - T_o}{\Sigma R}$, then the heat transfer rates across the system can be calculated.

Overall Heat Transfer Coefficient

2.3 Thermal Resistances, conductance and Overall Heat Transfer Coefficient

2.3.1 Heat Transfer in Plane Wall by Conduction

From Fourier's law of heat conduction in a plane wall shown in Fig. 2.3.1.

$$Q = -k \cdot A \cdot \frac{dT}{dx}$$

$$Q \cdot dx = -k \cdot A \cdot dT \quad \dots(i)$$

Boundary conditions are :

- At $x = 0$, $T = T_1$
- At $x = x$ or L , $T = T_2$

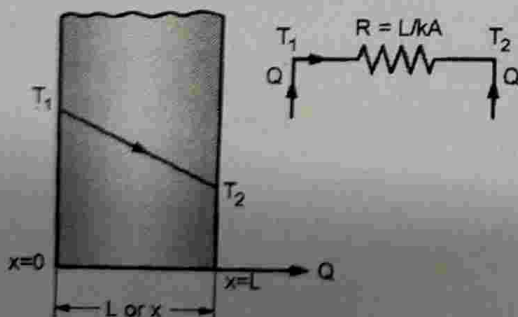


Fig. 2.3.1: Heat conduction in plane wall

On integrating the above equation within specified limits,

$$Q = \int_0^x dx = -kA \int_{T_1}^{T_2} dT$$

$$Q = \frac{k \cdot A (T_1 - T_2)}{x} = \frac{(T_1 - T_2)}{\left(\frac{x}{kA}\right)} = \frac{T_1 - T_2}{R} \quad \dots(2.3.1)$$

Where, Conductive resistance, $R = \frac{x}{kA} = \frac{L}{kA}$

2.3.2 Heat Transfer by Convection

Consider the heat transfer by convection from a surface to fluid in the direction normal to surface of area A as shown in Fig. 2.3.2.

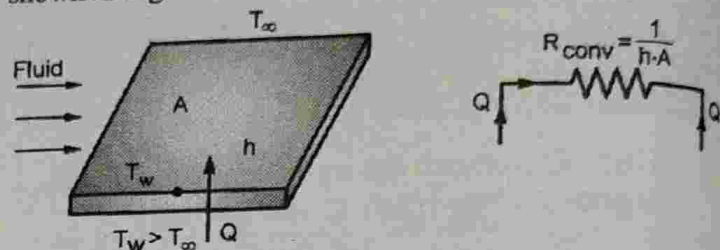


Fig. 2.3.2 : Heat transfer of convection

From Newton's law of cooling,

$$Q = h \cdot A \cdot (T_w - T_\infty) = \frac{(T_w - T_\infty)}{\frac{1}{hA}} = \frac{T_w - T_\infty}{R_{\text{convection}}} \quad \dots(2.3.2)$$

Where, convective thermal resistance,

$$R_{\text{convection}} = \frac{1}{h \cdot A} = \frac{T_w - T_\infty}{Q}$$

2.3.3 Overall Heat Transfer Coefficient, U

Heat flow rate through a composite wall and considering all modes of heat transfer (i.e. by conduction, convection and radiation) can be expressed as :

$$Q = U \cdot A (T_i - T_o) = \frac{(T_i - T_o)}{\frac{1}{UA}} = \frac{T_i - T_o}{\Sigma R} \quad \dots(2.3.3)$$

Where, T_i and T_o are temperatures at inner and outer sides of composite wall respectively and $T_i > T_o$.

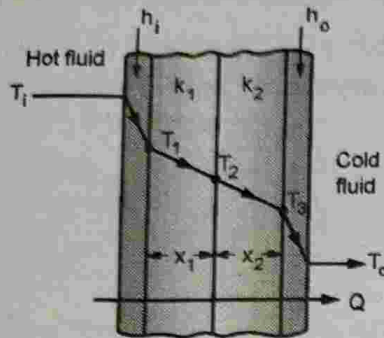
U = Overall heat transfer coefficient

$\Sigma R = \frac{1}{UA}$ = Overall or combined thermal resistance of the wall

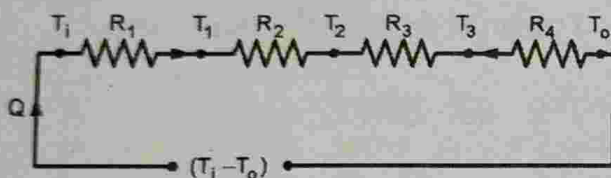
The overall heat transfer coefficient, U can be defined as the ability of a composite wall to transfer heat rate through it.

2.4 Heat Transfer Problem for Composite Wall having Resistances in Series

Consider a composite wall of thickness x_1 and x_2 of surface area A in the perpendicular direction to heat flow having hot fluid at temperature T_i on one side and cold fluid of temperature T_o on other side ($T_i > T_o$) as shown in Fig. 2.4.1(a).



(a) Heat transfer system



(b) Equivalent electrical system

Fig. 2.4.1 : Composite wall in series

Let h_i and h_o be the coefficient of heat transfer of films for hot fluid and cold fluid respectively.

The heat transfer from hot fluid to cold fluid is subjected to four resistances as shown in equivalent electrical system in Fig. 2.4.1(b).

$$\text{Where, } R_1 = \frac{1}{h_i A}, \quad R_2 = \frac{x_1}{k_1 A},$$

$$R_3 = \frac{x_2}{k_2 A}, \quad R_4 = \frac{1}{h_o A}$$

Total resistance,

$$\Sigma R = R_1 + R_2 + R_3 + R_4$$

$$\therefore Q = \frac{T_i - T_o}{\Sigma R} = \frac{T_i - T_o}{R_1 + R_2 + R_3 + R_4}$$

$$Q = \frac{T_i - T_o}{\frac{1}{h_i A} + \frac{x_1}{k_1 A} + \frac{x_2}{k_2 A} + \frac{1}{h_o A}} = \frac{T_i - T_o}{\frac{1}{UA}} \quad \dots(2.4.1)$$

$\therefore$ Overall heat transfer coefficient,

$$\frac{1}{U} = \frac{1}{h_i} + \frac{x_1}{k_1} + \frac{x_2}{k_2} + \frac{1}{h_o} \quad \dots(2.4.2)$$

2.4.1 Estimation of Intermediate Temperatures in the System

Since the heat transfer rate remains the same across the section under steady state, it follows that :

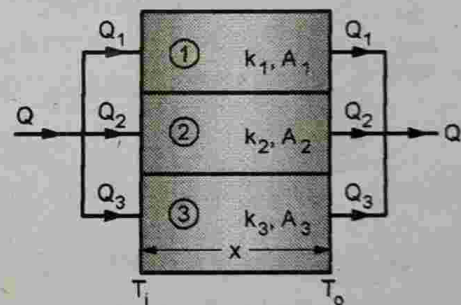
$$Q = h_i A (T_i - T_1) = \frac{k_1 A (T_1 - T_2)}{x_1} \\ = \frac{k_2 A (T_2 - T_3)}{x_2} = h_o A (T_3 - T_o) \quad \dots(2.4.3)$$

Using the above Equation (2.4.3), the intermediate temperature T_1 , T_2 and T_3 can be calculated.

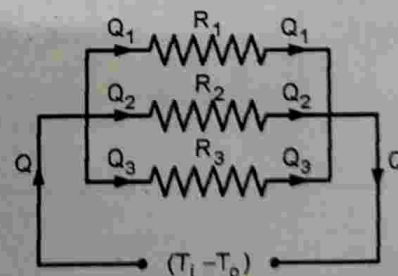
2.5 Heat Transfer Problem for Composite Slab having Resistance in Parallel

- (i) Neglecting Convective Heat Transfer
- (ii) Composite slab with convective Heat transfer (combined resistance in series and parallel)

2.5.1 Neglecting Convective Heat Transfer



(a) Heat transfer system



(b) Equivalent electrical system

Fig. 2.5.1 : Composite wall in parallel

Consider a heat transfer system as shown in Fig. 2.5.1(a) and its equivalent electrical system is shown in Fig. 2.5.1(b). Various resistances in the system are :

$$R_1 = \frac{x}{k_1 A_1}, \quad R_2 = \frac{x}{k_2 A_2}, \quad R_3 = \frac{x}{k_3 A_3}$$

Let Q_1, Q_2, Q_3 be the heat transfer rates in slabs 1, 2 and 3 respectively. Then,

$$Q_1 = \frac{T_1 - T_o}{R_1}, \quad Q_2 = \frac{T_1 - T_o}{R_2}, \quad Q_3 = \frac{T_1 - T_o}{R_3} \quad \dots(i)$$

But total heat transfer rate,

$$Q = Q_1 + Q_2 + Q_3 \quad \dots(ii)$$

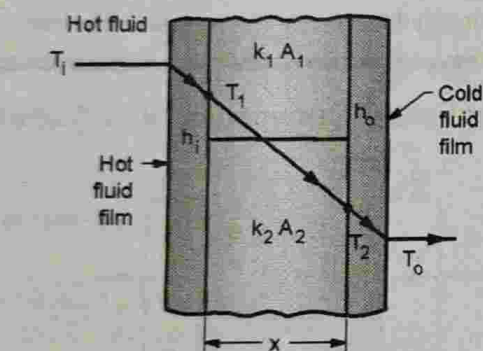
$$Q = \frac{T_1 - T_o}{\Sigma R} \quad \dots(iii)$$

Substituting the values from Equation (iii) and set of Equation (i) in Equation (ii), we get,

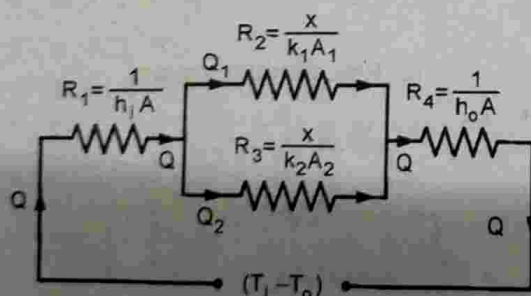
$$\frac{T_1 - T_o}{\Sigma R} = \frac{T_1 - T_o}{R_1} + \frac{T_1 - T_o}{R_2} + \frac{T_1 - T_o}{R_3}$$

$$\text{i.e. } \frac{1}{\Sigma R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \quad \dots(2.5.1)$$

2.5.2 Composite Slab with Convective Heat Transfer (Combined Resistance in Series and Parallel)



(a) Heat transfer system



(b) Equivalent electrical system

Fig. 2.5.2 : Composite slab with convective heat transfer

Refer Fig. 2.5.2. Various resistances are as follows :

$$R_1 = \frac{1}{h_1 A}, \quad R_2 = \frac{x}{k_1 A_1},$$

$$R_3 = \frac{x}{k_2 A_2}, \quad R_4 = \frac{1}{h_0 A}$$

Where, $A = A_1 + A_2$

Equations for heat flow rate are :

$$Q = \frac{T_1 - T_1}{R_1}, \quad Q_1 = \frac{T_1 - T_2}{R_2},$$

$$Q_2 = \frac{T_1 - T_2}{R_3}, \quad Q = \frac{T_2 - T_o}{R_4}$$

Combined resistance of the system can be calculated as follows :

For resistances in parallel,

$$\frac{1}{R_e} = \frac{1}{R_2} + \frac{1}{R_3} = \frac{R_2 + R_3}{R_2 \cdot R_3} \quad \dots(i)$$

$$\therefore \Sigma R = R_1 + R_e + R_4 \quad \dots(2.5.2)$$

$$Q = \frac{T_1 - T_o}{\Sigma R} \quad \dots(2.5.3)$$

Thermal Contact Resistance

2.6 Thermal Contact Resistance

- Consider one dimensional heat flow through a composite slab having two surfaces as shown in Fig. 2.6.1.
- The temperature profile through the solid surface at their interface experiences a sudden in temperature between two bodies as shown.

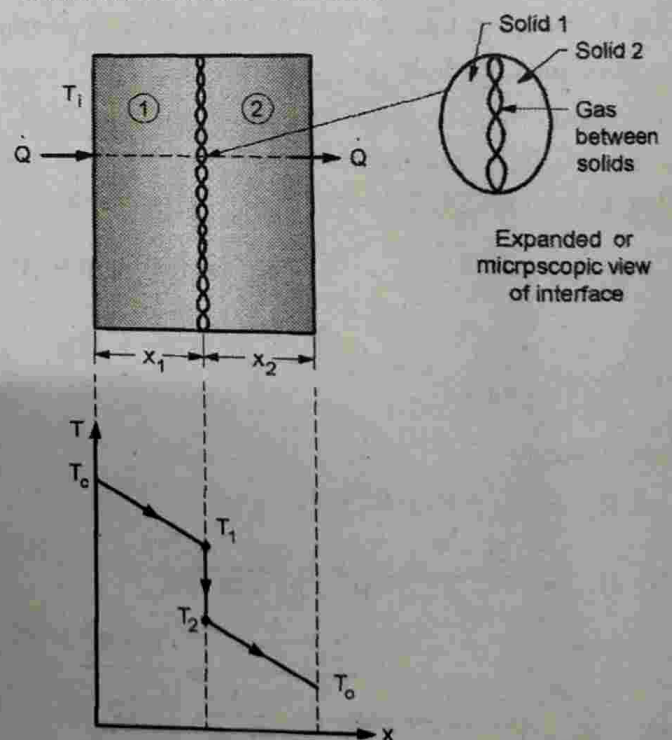


Fig. 2.6.1 : Contact resistance between two solid surfaces

- This temperature drop across the interface is as result of **thermal contact resistance**.
- This **thermal contact resistance** at the interface develops when the two surfaces do not fit tightly together and a thin layer of fluid (air or surrounding fluid) is filled between them.
- This contact resistance is the function of surface roughness, the pressure holding the surfaces, the property of fluid and the interface temperature.
- The expanded view of the interface is shown in Fig. 2.6.1. The direct contact between the solid surfaces takes place at a limited number of spots and the void is filled by the surrounding fluid.
- The heat conduction through the fluid filling the voids is mainly by conduction, since there is no convection in such a thin layer of fluid and the radiation effects are negligible.

Definition of Thermal Contact Resistance :

In case the thermal conductivity of fluid filled in voids is less than the thermal conductivity of the solids, the interface acts as a resistance to heat flow, called as thermal contact resistance.

2.7 Heat Transfer through a Composite Cylinder with Conduction and Convection

Refer Fig. 2.7.1.

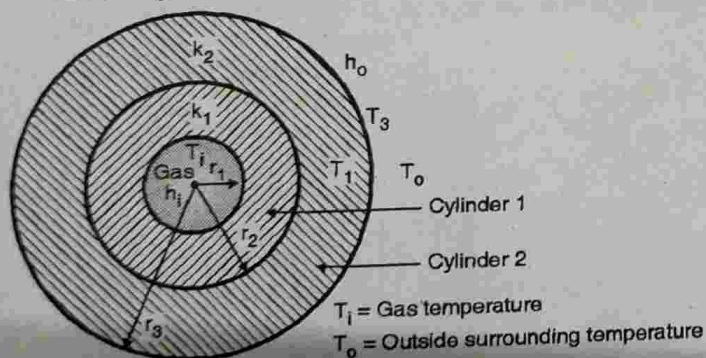


Fig. 2.7.1 : Heat transfer through a composite cylinder with conduction and convection

Heat transfer can be determined for a composite wall by considering the thermal resistance of convective heat transfer at inner layer, thermal conductance of cylinder 1 and 2 and convective heat transfer at outer layer of cylinder length L.

Resistance due to convection R_1 and R_4 and conduction R_2 and R_3 are as follows :

$$R_1 = \frac{1}{h_1 \cdot 2\pi r_1 L}, \quad R_2 = \frac{\log_e \left(\frac{r_2}{r_1} \right)}{2\pi k_1 L}$$

$$R_3 = \frac{\log_e \left(\frac{r_3}{r_2} \right)}{2\pi k_2 L}, \quad R_4 = \frac{1}{h_0 \cdot 2\pi r_3 L}$$

Heat transfer rate through various layers is as follows :

$$Q = h_1 \cdot 2\pi r_1 L (T_1 - T_2) = \frac{2\pi k_1 L (T_1 - T_2)}{\log_e \frac{r_2}{r_1}}$$

$$= \frac{2\pi k_2 L (T_2 - T_3)}{\log_e \frac{r_3}{r_2}}$$

$$= h_0 \cdot 2\pi r_3 L (T_3 - T_0) \quad \dots(2.7.1)$$

$$Q = \frac{(T_1 - T_0)}{R_1 + R_2 + R_3 + R_4}$$

$$= \frac{T_1 - T_0}{\frac{1}{h_1 \cdot 2\pi r_1 L} + \frac{\log_e \left(\frac{r_2}{r_1} \right)}{2\pi k_1 L} + \frac{\log_e \left(\frac{r_3}{r_2} \right)}{2\pi k_2 L} + \frac{1}{h_0 \cdot 2\pi r_3 L}} \quad \dots(2.7.2)$$

2.8 Combined Heat Transfer by Conduction and Convection in Compound Sphere

Refer Fig. 2.8.1.

Consider a hollow sphere having gas a temperature T_1 and outside temperature is T_0 . Let T_1 , T_2 , T_3 be the temperatures on inside surface, at interface of two spheres and T_3 at outer surface of sphere respectively.

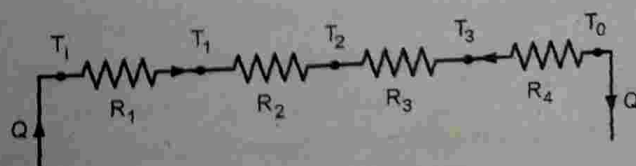
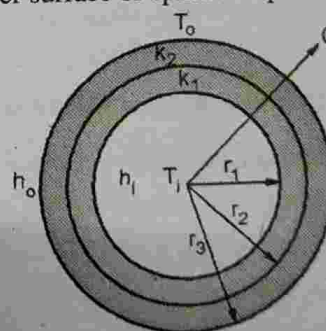


Fig. 2.8.1 : Compound sphere having heat transfer by conduction and convection

$$Q = h_1 \cdot A_1 \cdot (T_1 - T_2) = \frac{4\pi k_1 (T_1 - T_2)}{\left(\frac{1}{r_1} - \frac{1}{r_2}\right)}$$

$$= \frac{4\pi k_2 (T_2 - T_3)}{\left(\frac{1}{r_2} - \frac{1}{r_3}\right)} = h_o A_o (T_3 - T_o)$$

$$\text{or, } Q = \frac{T_1 - T_2}{\frac{1}{A_1 h_1}} = \frac{(T_1 - T_2)}{4\pi k_1 r_1 r_2}$$

$$= \frac{(T_2 - T_3)}{4\pi k_2 r_2 r_3} = \frac{T_3 - T_o}{h_o \cdot A_o} \quad \dots(2.8.1)$$

$$\text{where } A_1 = 4\pi r_1^2 \text{ and } A_o = 4\pi r_3^2$$

$$R_1 = \frac{1}{A_1 h_1}, \quad R_2 = \frac{(r_2 - r_1)}{4\pi k_1 \cdot r_1 \cdot r_2}$$

$$R_3 = \frac{(r_3 - r_2)}{4\pi k_2 \cdot r_2 \cdot r_3} \text{ and } R_4 = \frac{1}{A_o h_o}$$

$$\therefore Q = \frac{T_1 - T_o}{R_1 + R_2 + R_3 + R_4} \quad \dots(2.8.2)$$

Thermal Insulation & Critical Radius of Insulation

2.9 Thermal Insulation

- A heat insulating material is one which has low thermal conductivity.
- Thermal insulation is provided in thermal systems to reduce the heat losses

For example, the steam generated in the boiler is taken into pipes upto the steam turbine. To reduce the heat losses, the pipe is lagged with some form of insulating material.

Required properties of insulating materials are :

- It is able to withstand high or low temperatures.
- It should have long life and could withstand rough handling.
- It must be easy to apply.
- It is economical.
- It should not have any fire risks.

Various insulating materials used in practice are asbestos, glass and rock wool, cork, a man made plastic material like expanded polystyrene, magnesium carbonate etc.

Coefficient of thermal conductivity of various materials is given in Table 2.9.1.

Table 2.9.1 : Coefficient of thermal conductivity (k)

Substance	k (W/mK)	Substance	k (W/mK)
Asphalt	0.75	Marble	1.3
Fire brick	0.14	Cement	0.29
Glass wool	0.038	Glass	0.75
Mica	0.58	Asbestos	0.15
Gypsum	0.30	Cork	0.042
Brick	0.7	Fibre	0.049
Wood	1.1	Clay	1.03
Concrete	1.38	Saw dust	0.07
Plaster	0.80		

2.10 Critical Thickness of Insulation

Purpose of insulation is to reduce the heat transfer rate but is it always true ? Let us examine.

- Insulation in case of plane walls
- Application of insulation on cylinders

1. Insulation in case of plane walls

Consider the case of heat flow across a steel plate with a layer of insulation applied on it as shown in Fig. 2.10.1.

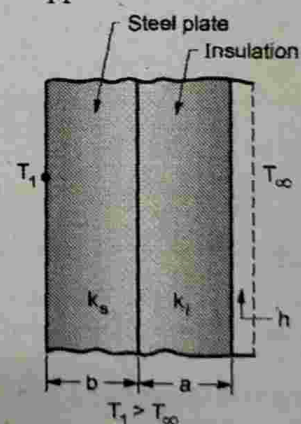


Fig. 2.10.1: Insulation on plane wall

Heat transfer rate without insulation,

$$Q = \frac{T_1 - T_\infty}{\left(\frac{b}{k_s A} + \frac{1}{h A}\right)} \quad \dots(i)$$

Heat transfer rate with insulation,

$$Q_1 = \frac{T_1 - T_\infty}{\frac{b}{k_s \cdot A} + \frac{a}{k_i \cdot A} + \frac{1}{h \cdot A}} \quad \dots(ii)$$

Due to increase in thermal resistance of insulation equal to $(a / k_i \cdot A)$, the value of Q_1 is always less than Q . It implies that the heat transfer rate will always reduce with insulation in case of plane walls.

2. Application of Insulation on cylinders

- Consider heat flow from steel tube of outside radius of r_1 . This is insulated by a layer of insulation so that outer radius of insulation is r_2 .
- Let the temperature of outside surface of steel tube be T_1 , conductivity of insulation be k (W/mK) and let this insulation be exposed to atmospheric air at temperature T_∞ with convective heat transfer coefficient as h (W/m²K) and length of tube L meters. Refer Fig. 2.10.2.

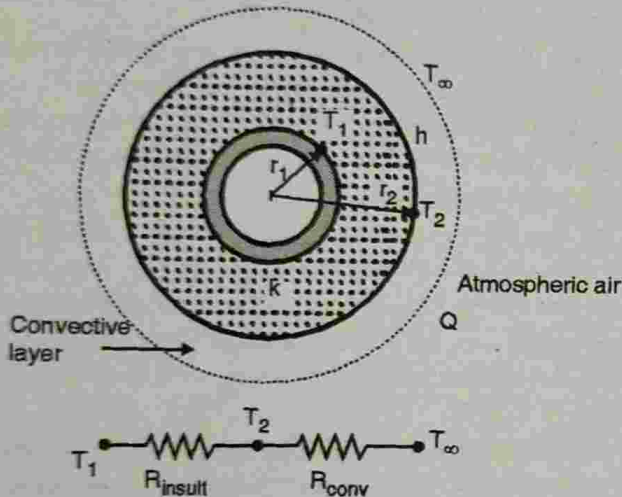


Fig. 2.10.2 : Critical thickness

Now heat transfer rate from this insulated steel tube,

$$Q = \frac{T_1 - T_\infty}{\frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k L} + \frac{1}{h \cdot 2\pi r_2 \cdot L}} \quad \dots(2.10.1)$$

- From the above equation it can be seen that on increase of insulation i.e. r_2 , heat flow rate Q may decrease or increase with increase in r_2 since conductive resistance

$\frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k L}$ increase logarithmically but convective resistance $\left(\frac{1}{h \cdot 2\pi r_2 L}\right)$ decreases linearly.

- If we plot Q as a function of r_2 , (Refer Fig. 2.10.3), we see that Q first increases with increase in r_2 and then decreases passing through a maximum value.

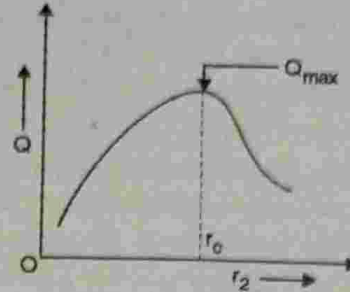


Fig. 2.10.3 : Variation of heat transfer w.r.t. insulation radius

- To find the value of r_2 , for which Q is maximum, $\frac{dQ}{dr_2}$ should be equated to zero or denominator of Equation (2.10.1) should be minimum, hence differentiating denominator with respect to r_2 and equating it to zero, we have

$$\begin{aligned} \therefore \frac{d}{dr_2} \left[\frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k L} + \frac{1}{h \cdot 2\pi r_2 \cdot L} \right] &= 0 \\ \text{or } \frac{d}{dr_2} \left[\frac{\ln\left(\frac{r_2}{r_1}\right)}{k} + \frac{1}{h r_2} \right] &= 0 \\ \therefore \frac{d}{dr_2} \left[\frac{\ln r_2}{k} - \frac{\ln r_1}{k} + \frac{1}{h r_2} \right] &= 0 \\ \frac{1}{r_2 \cdot k} - 0 - \frac{1}{h r_2^2} &= 0 \\ \frac{1}{k} - \frac{1}{h r_2} &= 0 \\ \frac{1}{k} &= \frac{1}{h r_2} \quad \text{or } r_2 = \frac{k}{h} = r_c \quad \dots(2.10.2) \end{aligned}$$

r_c is called critical radius of insulation.

2.10.1 Important Aspects of Critical Radius of Insulation

- With the increase in thickness of insulation (r_2), conductive resistance increases logarithmically and convective resistance decreases linearly, hence total resistance first decreases, attains minimum value (corresponding to maximum Q) and then increases.
- Critical radius is independent of tube/pipe radius. It only depends on conductivity of insulation and 'h' between exposed surface of insulation and its surroundings $\left(r_c = \frac{k}{h}\right)$.



- If the purpose of insulation is to reduce the heat transfer rate then outside radius of tube (r_1) should be greater than critical radius (r_c). **Example** : Steel tubes carrying steam from boiler to turbine.
- If purpose is to increase the heat transfer rate, then outer radius of tube (r_1) should be less than critical radius (r_c). **Example** : Insulation over current carrying electric conductors/cables.
- Heat transfer will always reduce on application of insulation on wall or similar structure as conductive resistance will increase with thickness of insulation while convective resistance will remain unchanged. Hence aspect of critical radius is not applicable.

2.11 Critical Thickness of Insulation in case of Spheres

Referring to Fig. 2.11.1,

Consider of hollow sphere at outer radius r_1 at temperature T_1 which is covered with an insulation of thickness ($r_2 - r_1$) so that its outer radius is r_2 .

It is subjected to convective heat transfer h having fluid temperature T_∞ . Let k be the thermal conductivity of insulation.

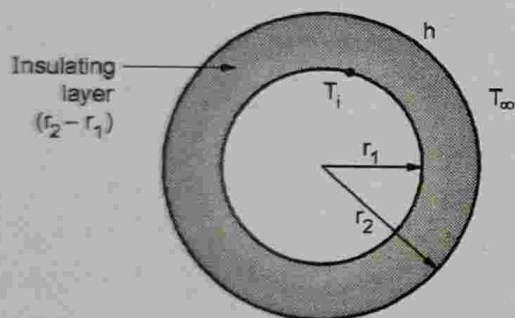


Fig. 2.11.1 : Sphere with insulation

Heat transfer rate across the sphere can be expressed as :

$$Q = \frac{(T_1 - T_\infty)}{\frac{(r_2 - r_1)}{4\pi k r_1 r_2} + \frac{1}{4\pi r_2^2 h}} \quad \dots(2.11.1)$$

Where, conductive resistance due to insulation,

$$R_1 = \frac{r_2 - r_1}{4\pi k r_1 r_2}$$

Convective resistance to atmosphere,

$$R_2 = \frac{1}{4\pi r_2^2 h}$$

For maximum heat transfer rate, $\frac{dQ}{dr_2} = 0$

$$\text{Therefore, } \frac{d}{dr_2} \left[\frac{T_1 - T_\infty}{\frac{(r_2 - r_1)}{4\pi k r_1 r_2} + \frac{1}{4\pi r_2^2 h}} \right] = 0$$

$$\frac{d}{dr_2} \left[\frac{T_1 - T_\infty}{\frac{1}{4\pi k} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) + \frac{1}{4\pi r_2^2 h}} \right] = 0$$

Differentiating the denominator for its minimum value,

$$\frac{d}{dr_2} \left[\frac{1}{k r_1} - \frac{1}{k r_2} + \frac{1}{r_2^2 h} \right] = 0$$

$$\frac{1}{k r_2^2} - \frac{2}{r_2^3 h} = 0$$

$$\therefore r_2 = \frac{2k}{h} = r_c \quad \dots(2.11.2)$$

where, r_c is called the critical radius of insulation.

2.12 Economic Thickness of Insulation

- Economic thickness of insulation is based on the net operating cost which is the sum of fixed operating cost and the production cost.
- When the insulation is provided for certain amount, The money is borrowed. *The cost of depreciation and its cost of maintenance is called the fixed operating cost.* This cost will obviously increase with increase in thickness of insulation.
- As the thickness of insulation is increased, the heat losses are reduced. It is corresponding to saving in cost of heat energy loss. It is called the **production cost**.
- A graph can be plotted between the cost taken on Y-axis and thickness of insulation on X-axis as shown in Fig. 2.12.1.
- As discussed above, the fixed operating cost is almost linear which increases with increase in thickness of insulation while the production cost decreases.

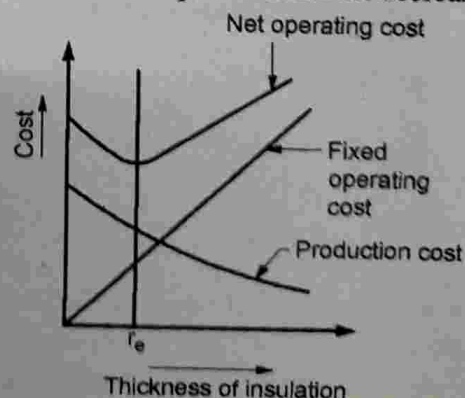


Fig. 2.12.1 : Economic thickness of insulation

- If we superimpose these two costs we get the net operating cost as shown in Fig. 2.12.1.

- The thickness of insulation, at which the net operating cost is minimum, is known as **Economic thickness of insulation** r_e .

2.12.1 Pay Back Period

- The simple pay back period represents the time required in number of months years to recover the initial investment (first cost i.e. the capital cost) by considering the net annual savings. It is calculated as follows :

- Simple Pay Back period (SPP) in year,

$$SPP = \frac{\text{First Cost or Capital Cost}}{\text{Yearly Benefits} - \text{Yearly Costs}} \quad \dots(2.12.1)$$

Example

If the capital cost is Rs. 100000/- and the yearly savings are at the rate of 25000/-. Then the simple pay back period equals to $\frac{100000}{25000} = 4$ years.

Smaller the payback period, more desirable is the project proposal.

Advantages

- It is simple in concept and easy to apply to various project proposals. Shorter the pay back period better is the proposal.
- This concept is useful to energy managers. Note that this concept is a measure of projects capital recovery and not the profitability.
- This concept is useful in selecting the best investment out of the various proposals in respect of energy efficient technology, maintenance and other charges.
- It favours those projects which generate more cash flow in the earlier years of investment as compared to projects which generate more cash flows in later years.

Limitations

- It fails to consider the time value of money.
- It ignores the cash flow beyond the pay back period.

Explanation of time value of money

Cash inflows in the payback calculations are simply added without discounting. However the principle of financial analysis requires that the cash flows occurring at different times can be added or subtracted only after suitable compounding or discounting.

Concept of Time value of money is referred to two evaluations i.e.

- Present value (PV) of future money.
 - Future value (FV) of present money
- Let i = Rate of annual interest in percentage
 n = Number of years in future

Then the relation between FV and NPV is given by the equation.

$$FV = NPV \left(1 + \frac{i}{100}\right)^n \quad \dots(2.12.2)$$

e.g. Rs 1000 received one year from now at (FV) at 10% interest will have its present value as :

$$NPV = \frac{FV}{\left(1 + \frac{i}{100}\right)^n} = \frac{1000}{\left(1 + \frac{10}{100}\right) \times 1} = \text{Rs. } 909.09 \text{ only.}$$

Table 2.12.1 : List of Formulae

- $k = k_0 (1 + \alpha \cdot T)$ and

$$\frac{Q}{A} \int_0^x dx = - \int_{T_1}^{T_2} k_0 (1 + \alpha T) \cdot dT$$
- $Q = \frac{\Delta T}{\Sigma R}$
- Conductive resistance, $R = \frac{x}{kA}$ or $\frac{L}{kA}$
- Convective resistance, $R_{conv} = \frac{1}{h \cdot A}$
- Overall heat transfer coefficient,

$$U = \frac{1}{A \cdot \Sigma R}$$

Where $Q = U \cdot A \cdot (T_i - T_o)$
- Resistance in series :
 $\Sigma R = R_1 + R_2 + R_3 + R_4$; Where,
 $R_1 = \frac{1}{h_i A}$; $R_2 = \frac{x_1}{k_1 A}$; $R_3 = \frac{x_2}{k_2 A}$; $R_4 = \frac{1}{h_o A}$
- Resistances in parallel (without convective resistance)
 $Q = Q_1 + Q_2 + Q_3$

$$= \frac{(T_i - T_o)}{R_1} + \frac{(T_i - T_o)}{R_2} + \frac{(T_i - T_o)}{R_3} = \frac{T_i - T_o}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$
- Heat transfer in composition cylinder

$$Q = \frac{T_i - T_o}{\frac{1}{h_i 2\pi r_1 L} + \frac{\ln(r_2/r_1)}{2\pi k_1 L} + \frac{\ln(r_3/r_2)}{2\pi k_2 L} + \frac{1}{h_o 2\pi r_2 L}}$$
- Heat transfer in composite sphere,

$$Q = \frac{T_1 - T_2}{\left(\frac{1}{A_i \cdot h_i}\right) + \left(\frac{(T_2 - T_3)}{4\pi k_1 r_1 r_2}\right) + \left(\frac{(T_3 - T_4)}{4\pi k_2 r_2 r_3}\right) + \frac{(T_4 - T_0)}{A_0 \cdot h_0}}$$

Where $A_i = 4\pi r_1^2$ and $A_0 = 4\pi r_2^2$

$$Q = \frac{T_1 - T_0}{R_1 + R_2 + R_3 + R_4}$$

10. Critical thickness of insulation

(a) In plane walls : Heat loss always reduces due to additional thermal resistance of insulation, hence, no critical thickness.

(b) Long cylinders : $r_2 = r_c = \frac{k}{h}$ = critical radius.

To reduce heat losses : $r_2 > r_c$

To increase heat transfer rates : $r_2 < r_c$

(c) Spheres : $r_2 = r_c = \frac{2k}{h}$

Ex. 2.12.1 : A furnace wall lining is made up of material with $k = 2.5 \text{ W/mK}$. The temperature of the inner and outer surface of this plane wall lining are 810°C and 330°C respectively. The outer surface is exposed to ambient air at 30°C with convective heat transfer coefficient = $10 \text{ W/m}^2\text{K}$. Calculate :

- The rate of heat flow per unit area.
- Thickness of lining in given situation.
- Thickness of lining if the heat flow rate is to be reduced by 50%.

Soln. : Refer Fig. P. 2.12.1.

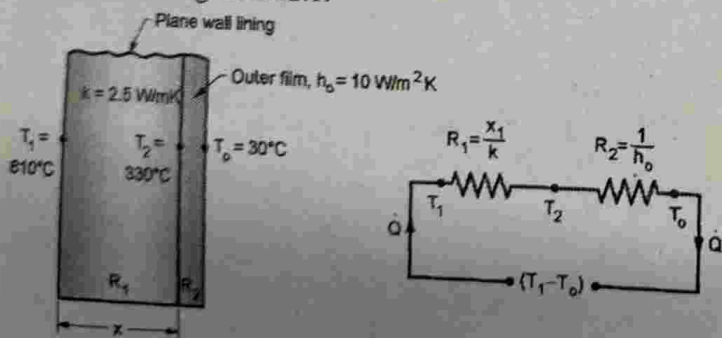


Fig. P. 2.12.1

$$R_1 = \frac{x_1}{k} = \frac{x_1}{2.5} \quad \text{and} \quad R_2 = \frac{1}{h_o} = \frac{1}{10} = 0.1 \text{ K/W}$$

(i) Rate of heat flow/unit area, $q = \frac{Q}{A}$

Since $q = \frac{(T_1 - T_2)}{R_1} = \frac{(T_2 - T_o)}{R_2} = \frac{(T_1 - T_o)}{R_1 + R_2}$

$$q = \frac{T_2 - T_o}{R_2} = \frac{(330 - 30)}{0.1} = 3000 \text{ W/m}^2$$

...Ans.

(ii) Thickness of the existing lining, x

$$q = \frac{T_1 - T_2}{\frac{x_1}{2.5} + 0.1}$$

i.e. $3000 = \frac{810 - 30}{\frac{x_1}{2.5} + 0.1}$

$$\therefore x_1 = 0.4 \text{ m or } 40 \text{ cm}$$

...Ans.

(iii) Thickness of lining x_1 required if the heat flow rate is reduced to 50% i.e. $q_1 = \frac{1}{2} q$

$$q_1 = \frac{(T_1 - T_o)}{R_1 + R_2} \quad \text{i.e.} \quad \frac{q}{2} = \frac{(T_1 - T_o)}{\frac{x_1}{k_1} + \frac{1}{h_o}}$$

$$\frac{3000}{2} = \frac{(810 - 30)}{\frac{x_1}{2.5} + \frac{1}{10}}$$

$$x_1 = 1.05 \text{ m}$$

...Ans.

Ex. 2.12.2 : A rectangular slab ($k = 10 \text{ W/mK}$) of thickness 15 cm and inside temperature of 400°C is insulated by a materials of thickness 10 cm ($k_i = 30 \text{ W/mK}$). The ambient air is at 28°C and the outside convective heat transfer coefficient is $15 \text{ W/m}^2\text{K}$. Determine the steady state heat transfer per unit surface area and the temperature of outside surface of the slab and the insulation.

Soln. :

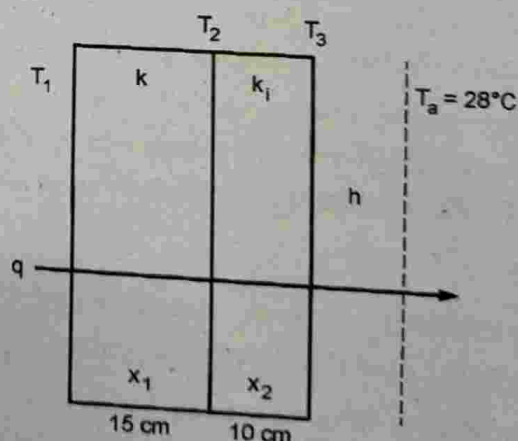


Fig. P. 2.12.2

Refer Fig. P. 2.12.2

$$T_1 = 400^\circ\text{C};$$

$$T_a = 28^\circ\text{C}$$

$$k = 10 \text{ W/mK};$$

$$k_i = 30 \text{ W/mK}$$

$$h = 15 \text{ W/m}^2\text{K}$$

(i) Heat transfer rate/unit surface area, q

$$q = \frac{T_1 - T_a}{\frac{x_1}{k} + \frac{x_2}{k_1} + \frac{1}{h}} = \frac{400 - 28}{\frac{0.15}{10} + \frac{0.10}{30} + \frac{1}{15}}$$

$$= \frac{372}{0.015 + 0.0023 + 0.0657}$$

$$= 4376.5 \text{ W/m}^2$$

...Ans.

(ii) Temperature of insulation, T_2

$$q = \frac{T_1 - T_2}{\frac{x_1}{k}} ; 4376.5 = \frac{400 - T_2}{(0.15/10)}$$

$$T_2 = 334.35^\circ\text{C}$$

...Ans.

(iii) Temperature of outside surface, T_3

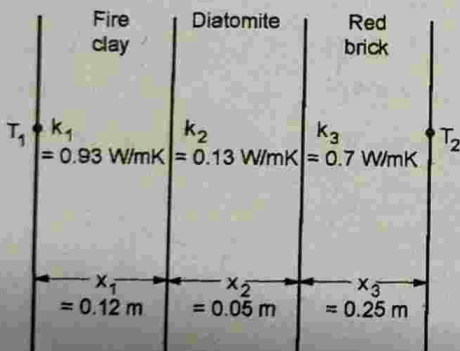
$$q = \frac{T_2 - T_3}{\frac{x_2}{k_1}} ; 4376.5 = \frac{334.35 - T_3}{(0.1/30)}$$

$$T_3 = 319.9^\circ\text{C}$$

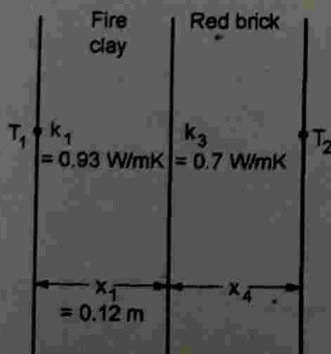
...Ans.

Ex. 2.12.3 : A brick work of a furnace is built up of layers laid of fire clay ($k_1 = 0.93 \text{ W/mK}$) and red brick ($k_3 = 0.7 \text{ W/mK}$) and the space between the two is filled with crushed diatomite brick ($k_2 = 0.13 \text{ W/mK}$) as shown in Fig. P. 2.12.3. What should be the thickness brick layer if the brick work is to be laid without diatomite filling between two layers so that the heat flow rate through the work remains constant with same temperatures of the furnace and outside temperatures.

Soln. :



(a) Brickwork with diatomite filling



(b) Brickwork without diatomite filling

Fig. P. 2.12.3

Case I : When diatomite filling is used

[Refer Fig. P. 2.12.3(a)]

Let A be surface area of composite wall with T_1 as furnace temperatures and T_2 surrounding temperature.

Combined thermal resistance,

$$\begin{aligned} \Sigma R &= R_1 + R_2 + R_3 \\ &= \frac{x_1}{k_1 A} + \frac{x_2}{k_2 A} + \frac{x_3}{k_3 A} \\ &= \frac{1}{A} \left[\frac{0.12}{0.93} + \frac{0.05}{0.13} + \frac{0.25}{0.7} \right] = \frac{0.8708}{A} \end{aligned} \quad \dots(i)$$

Case II : When diatomite is not used, let x_4 be the thickness of red brick as shown in Fig. P. 2.12.3(b). New combined resistance,

$$\begin{aligned} \Sigma R_0 &= R_1 + R_4 = \frac{x_1}{k_1 A} + \frac{x_4}{k_3 A} = \frac{1}{A} \left[\frac{0.12}{0.93} + \frac{x_4}{0.6} \right] \\ &= \frac{1}{A} [0.129 + 1.4286 x_4] \end{aligned} \quad \dots(ii)$$

For the same heat flow rate, it implies that

$$Q = \frac{T_1 - T_2}{\Sigma R} = \frac{T_1 - T_2}{\Sigma R_0}$$

$$\text{i.e. } \Sigma R = \Sigma R_0$$

$$\frac{0.8708}{A} = \frac{1}{A} [0.129 + 1.4286 x_4]$$

$$x_4 = 0.5193 \text{ m or } 51.93 \text{ cm}$$

...Ans.

Ex. 2.12.4 : An exterior wall of a house may be approximated by a 100 mm layer of common brick ($k_1 = 0.7 \text{ W/mK}$) followed by a 40 mm layer of gypsum plaster ($k_2 = 0.065 \text{ W/mK}$). What thickness of loosely packed rock-wool insulation ($k_3 = 0.065 \text{ W/mK}$) should be added to reduce the loss (or gain) through wall by 80%.

Soln. : Refer Fig. P. 2.12.4.

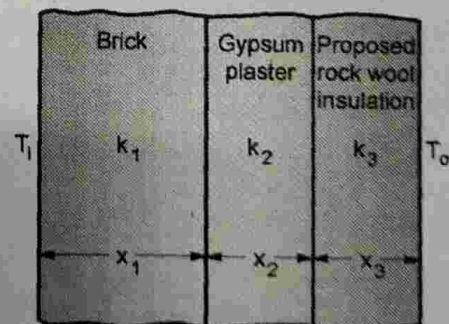


Fig. P. 2.12.4

$$x_1 = 100 \text{ mm} = 0.1 \text{ m}$$

$$k_1 = 0.7 \text{ W/mK}$$

$$x_2 = 40 \text{ mm} = 0.04 \text{ m}$$

$$k_2 = 0.065 \text{ W/mK}$$

$$k_3 = 0.065 \text{ W/mK}$$

$$q_2 = (1 - 0.8) q_1 = 0.2 \times q_1 \quad \dots(i)$$

(Since 80% i.e. 0.8 times the heat loss or gain is to be reduced). Consider $(T_i - T_o)$ be the temperature difference across the wall. The rate of heat flux q_1 without rockwool insulation and q_2 with rockwool insulation can be written as (where, $q = \frac{Q}{A}$)

$$q_1 = \frac{(T_i - T_o)}{\frac{x_1}{k_1} + \frac{x_2}{k_2}} = \frac{(T_i - T_o)}{\left(\frac{0.1}{0.7} + \frac{0.04}{0.065}\right)} = \frac{(T_i - T_o)}{0.7582} \quad \dots(ii)$$

$$q_2 = \frac{(T_i - T_o)}{\frac{x_1}{k_1} + \frac{x_2}{k_2} + \frac{x_3}{k_3}} = \frac{(T_i - T_o)}{\frac{0.1}{0.7} + \frac{0.04}{0.065} + \frac{x_3}{0.065}} \quad \dots(iii)$$

$$= \frac{T_i - T_o}{(0.7582 + 15.384 x_3)}$$

$$\text{Since } q_2 = 0.2 q_1$$

$$\therefore \frac{(T_i - T_o)}{0.7582 + 15.384 x_3} = \frac{0.2 \times (T_i - T_o)}{0.7582}$$

$$0.7582 = 0.2 \times 0.7582 + 0.2 \times 15.384 x_3$$

$$x_3 = 0.1971 \text{ m} = 19.71 \text{ cm} \quad \dots\text{Ans.}$$

Ex. 2.12.5 : A 8 mm thick metal plate, having thermal conductivity ($k = 98.6 \text{ W/mK}$) is exposed to vapour at 100°C on one side and cooling water at 30°C on opposite side. The heat transfer coefficients are $h_i = 14200 \text{ W/m}^2\text{K}$ on vapour side and $h_o = 2325 \text{ W/m}^2\text{K}$ on the water side. Determine the rate of heat transfer, drop in temperature of each side of plate.

Soln. : System is shown in Fig. P. 2.12.5.

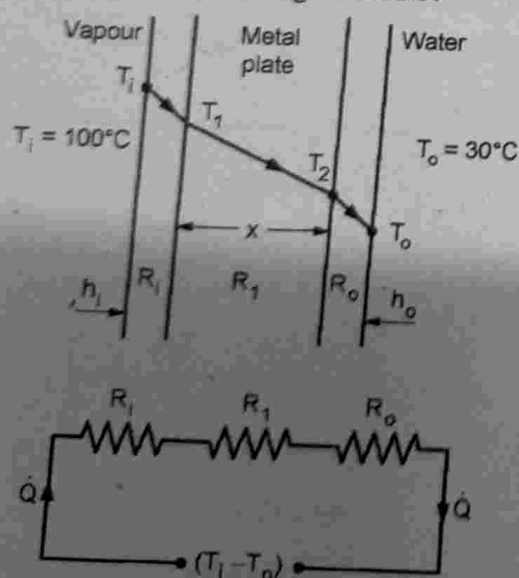


Fig. P. 2.12.5

$$x = 8 \text{ mm} = 8 \times 10^{-2} \text{ m} = 0.008 \text{ m}$$

$$k = 98.6 \text{ W/mK}, \quad T_i = 100^\circ\text{C}$$

$$T_o = 30^\circ\text{C},$$

$$h_o = 2325 \text{ W/m}^2\text{K}$$

$$h_i = 14200 \text{ W/m}^2\text{K}$$

Consider unit area of surface i.e. $A = 1 \text{ m}^2$ various

resistances are :

$$R_i = \frac{1}{h_i A} = \frac{1}{14200 \times 1} = 7.042 \times 10^{-5} \text{ K/W}$$

$$R_1 = \frac{x}{kA} = \frac{0.008}{98.6 \times 1} = 8.114 \times 10^{-5} \text{ K/W}$$

$$R_o = \frac{1}{h_o A} = \frac{1}{2325 \times 1} = 43.011 \times 10^{-5} \text{ K/W}$$

Combined resistance,

$$\begin{aligned} \Sigma R &= R_i + R_1 + R_o \\ &= (7.042 \times 10^{-5} + 8.114 \times 10^{-5} + 42.011 \times 10^{-5}) \\ &= 58.167 \times 10^{-5} \text{ K/W} \end{aligned}$$

(i) Heat transfer rate Q

$$Q = \frac{T_i - T_o}{\Sigma R} = \frac{(100 - 30)}{58.167 \times 10^{-5}} = 1.2034 \times 10^5 \text{ W} \quad \dots\text{Ans.}$$

(ii) Drop in temperature on each side of heat transfer

(a) Temperature drop on vapour side $(T_i - T_1)$:

$$Q = \frac{T_i - T_1}{R_i} ;$$

$$(T_i - T_1) = Q \times R_i$$

$$(T_i - T_1) = (1.2034 \times 10^5) \times (7.042 \times 10^{-5}) = 8.47^\circ\text{C} \quad \dots\text{Ans.}$$

(b) Temperature drop on cooling water side $(T_2 - T_o)$

$$Q = \frac{T_2 - T_o}{R_o}$$

$$\begin{aligned} \therefore (T_2 - T_o) &= Q \times R_o = (1.2034 \times 10^5) \times (43.011 \times 10^{-5}) \\ &= 51.76^\circ\text{C} \quad \dots\text{Ans.} \end{aligned}$$

Ex. 2.12.6 : A furnace wall is made up of three layers of thickness 250 mm, 100 mm and 150 mm having coefficient of thermal conductivities of 1.65 W/mK, 9.2 W/mK and k_3 (W/m K) respectively. The inside surface is exposed to hot gases of furnace at 1250°C with convective coefficient of heat transfer of $25 \text{ W/m}^2\text{K}$ and the inside surface temperature is 1100°C . The outside surface of composite wall is exposed to air at 25°C having coefficient of convective heat transfer of $12 \text{ W/m}^2\text{K}$.

Determine the following :

- Heat transfer rate per unit surface area.
- Thermal conductivity of third layer, k_3
- Overall heat transfer coefficient, U
- Intermediate surface temperatures.

Soln. :

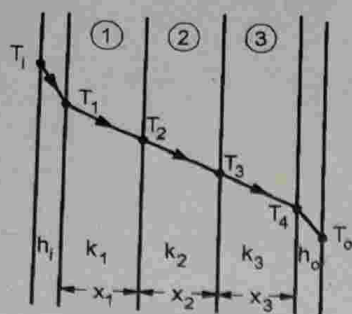


Fig. P. 2.12.6

Refer Fig. P. 2.12.6.

$$\begin{aligned}
 A &= 1 \text{ m}^2, & T_i &= 1250^\circ\text{C}; \\
 T_1 &= 1100^\circ\text{C}; & T_o &= 25^\circ\text{C} \\
 x_1 &= 250 \text{ mm} = 0.25 \text{ m} \\
 x_2 &= 100 \text{ mm} = 0.1 \text{ m} \\
 x_3 &= 150 \text{ mm} = 0.15 \text{ m} \\
 k_1 &= 1.65 \text{ W/mK}; & k_2 &= 9.2 \text{ W/mK}; \\
 h_i &= 25 \text{ W/m}^2\text{K}; \\
 h_o &= 12 \text{ W/m}^2\text{K}.
 \end{aligned}$$

(i) Heat transfer rate, Q

$$\begin{aligned}
 Q &= h_i \cdot A (T_i - T_1) \\
 &= 25 \times 1 (1250 - 1100) = 3750 \text{ W/m}^2 \dots \text{Ans.}
 \end{aligned}$$

(ii) Thermal conductivity of third layer, k_3

Total thermal resistance,

$$\begin{aligned}
 \sum R &= \frac{1}{h_i A} + \frac{x_1}{k_1 A} + \frac{x_2}{k_2 A} + \frac{x_3}{k_3 A} + \frac{1}{h_o A} \\
 &= \frac{1}{25 \times 1} + \frac{0.25}{1.65 \times 1} + \frac{0.1}{9.2 \times 1} + \frac{0.15}{k_3 \times 1} + \frac{1}{12 \times 1} \\
 &= 0.2857 + \frac{0.15}{k_3} = \frac{0.2857 k_3 + 0.15}{k_3} \dots (i)
 \end{aligned}$$

$$Q = \frac{T_i - T_o}{\sum R}$$

$$\text{i.e. } 3750 = \frac{(1250 - 25)}{\left(\frac{0.2857 k_3 + 0.15}{k_3} \right)}$$

On solving,

$$k_3 = 3.662 \text{ W/mK} \dots \text{Ans.}$$

(iii) Overall heat transfer coefficient, U

$$\begin{aligned}
 Q &= U \cdot A (T_i - T_o) \\
 3750 &= U \times 1 (1250 - 25)
 \end{aligned}$$

$$U = 3.0612 \text{ W/m}^2\text{K}$$

...Ans.

(iv) Surface temperatures T_2 , T_3 and T_4

For layer 1 :

$$\begin{aligned}
 Q &= \frac{k_1 A (T_1 - T_2)}{x_1} \\
 3750 &= \frac{1.65 \times 1 \times (1100 - T_2)}{0.25}
 \end{aligned}$$

$$\therefore T_2 = 531.82^\circ\text{C}$$

...Ans.

For layer 2 :

$$\begin{aligned}
 Q &= \frac{k_2 A (T_2 - T_3)}{x_2} \\
 3750 &= \frac{9.2 \times 1 \times (531.82 - T_3)}{0.1}
 \end{aligned}$$

$$T_3 = 491.06^\circ\text{C}$$

...Ans.

For layer 3 : $Q = \frac{k_3 A (T_3 - T_4)}{x_3}$

$$3750 = \frac{3.662 \times 1 \times (491.06 - T_4)}{0.15}$$

$$T_4 = 337.46^\circ\text{C}$$

...Ans.

Ex. 2.12.7 : The wall of a cold room has three layers. Outside layer is made of brick of 200 mm thickness. Middle layer is of cork 100 mm thick. Inside layer is of cement 50 mm thick. Outside air is at 25°C and inside air is at -20°C . Film coefficient of outside air/brick is $45.4 \text{ W/m}^2\text{C}$ and inside film coefficient for air /Cement is $17 \text{ W/m}^2\text{C}$. Values for 'k' for brick, cork and cement are 3.45, 0.043 and 0.294 W/mC respectively.

- Sketch the cross-section of composite wall with temperature profile and an analogous electrical circuit.
- Find the rate of steady flow under steady state conditions.
- Determine the temperature on exposed wall surfaces.

Soln. :

(i) Sketch of cross-section of composite wall

It is shown in Fig. P. 2.12.7(a) and equivalent electrical system in Fig. P. 2.12.7(b).

(ii) Heat flow rate, Q Consider unit surface area of wall i.e. $A = 1 \text{ m}^2$.

Resistances of various surfaces per unit area are :

$$R_1 = \frac{1}{h_i A} = \frac{1}{17 \times 1} = 0.05882 \text{ K/W}$$



$$R_2 = \frac{x_1}{k_1 A} = \frac{(50/1000)}{0.294 \times 1} = 0.17007 \text{ K/W}$$

$$R_3 = \frac{x_2}{k_2 A} = \frac{(100/1000)}{0.043 \times 1} = 2.32558 \text{ K/W}$$

$$R_4 = \frac{x_3}{k_3 A} = \frac{(200/1000)}{3.45 \times 1} = 0.05797 \text{ K/W}$$

$$R_5 = \frac{1}{h_o A} = \frac{1}{45.4 \times 1} = 0.02203$$

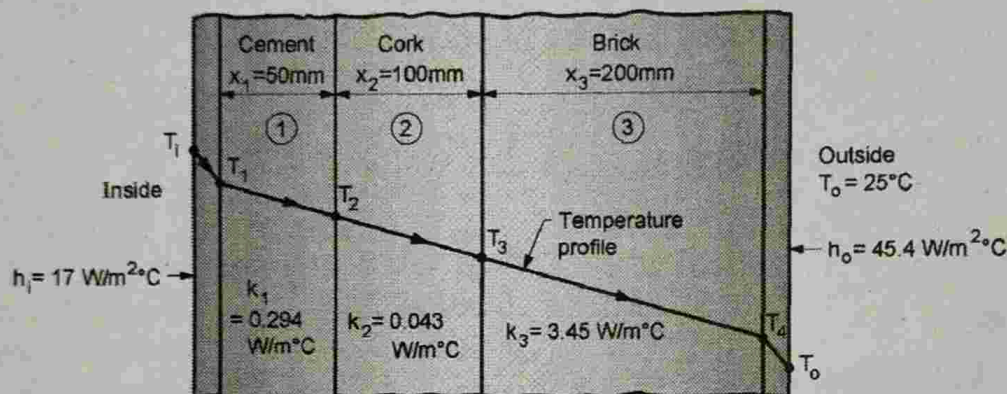
Combined resistance,

$$\begin{aligned} \Sigma R &= R_1 + R_2 + R_3 + R_4 + R_5 \\ &= 0.05882 + 0.17007 + 2.32558 + 0.05797 + 0.02203 = 2.63447 \text{ K/W} \end{aligned}$$

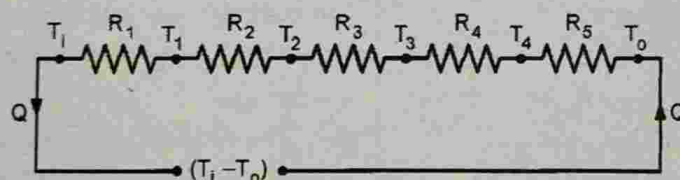
$$Q = \frac{(T_i - T_o)}{\Sigma R} = \frac{-20 - 25}{2.63447} = -17.08 \text{ W}$$

...Ans.

Negative sign shows that heat will flow from outside to inside. The direction of heat flow is shown in Fig. P. 2.12.7(b).



(a) Heat transfer system



(b) Equivalent electrical system

Fig. P. 2.12.7

(iii) Temperatures on exposed inside wall (cement) surface, T_1 and on exposed outside wall (brick) surface, T_4

$$Q = \frac{T_i - T_1}{R_1};$$

$$-17.08 = \frac{-20 - T_1}{0.05882}$$

$$\therefore T_1 = -18.995^\circ\text{C}$$

...Ans.

$$\text{Also, } Q = \frac{T_4 - T_o}{R_5}$$

$$-17.08 = \frac{T_4 - 25}{0.02203}$$

$$\therefore T_4 = 24.624^\circ\text{C}$$

...Ans.

Ex. 2.12.8 : Calculate the heat transfer rate through a composite wall shown in Fig. P. 2.12.8(a). Given that :

$$k_1 = 40 \text{ W/mK},$$

$$k_2 = 10 \text{ W/mK},$$

$$k_3 = 6 \text{ W/mK},$$

$$k_4 = 25 \text{ W/mK}$$

$$x_1 = 0.1 \text{ m},$$

$$x_2 = x_3 = 0.2 \text{ m}, x_4 = 0.06 \text{ m}$$

$$T_1 = 1000^\circ\text{C}$$

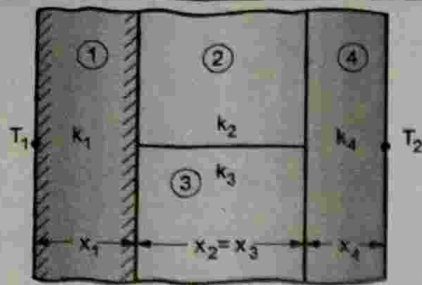
$$T_2 = 200^\circ\text{C}, A_1 = 1 \text{ m}^2,$$

$$A_2 = A_3 = 0.5 \text{ m}^2,$$

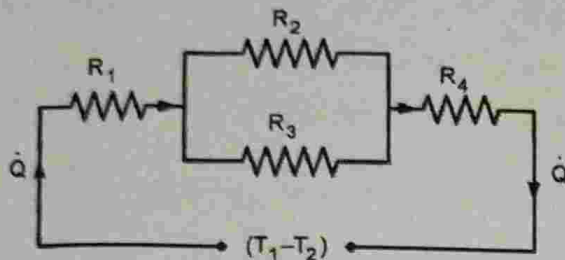
$$A_4 = 1 \text{ m}^2.$$

Also draw equivalent electrical system.

Soln. : Heat transfer system and equivalent electrical system are represented in Fig. P. 2.12.8.



(a) Heat transfer system

(b) Equivalent electrical system
Fig. P. 2.12.8

Resistances of various walls are :

$$R_1 = \frac{x_1}{k_1 A_1} = \frac{0.1}{40 \times 1} = 0.0025 \text{ K/W}$$

$$R_2 = \frac{x_2}{k_2 A_2} = \frac{0.2}{10 \times 0.5} = 0.04 \text{ K/W}$$

$$R_3 = \frac{x_3}{k_3 A_3} = \frac{0.2}{6 \times 0.5} = 0.0067 \text{ K/W}$$

$$R_4 = \frac{x_4}{k_4 A_4} = \frac{0.06}{25 \times 1} = 0.0024 \text{ K/W}$$

Equivalent resistance of loop having resistances in parallel is,

$$\frac{1}{R_{eq}} = \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{0.04} + \frac{1}{0.0067}$$

$$R_{eq} = 0.00574 \text{ K/W}$$

Combined resistance,

$$\begin{aligned} \sum R &= R_1 + R_{eq} + R_4 \\ &= 0.0025 + 0.00574 + 0.0024 = 0.01064 \text{ K/W} \end{aligned}$$

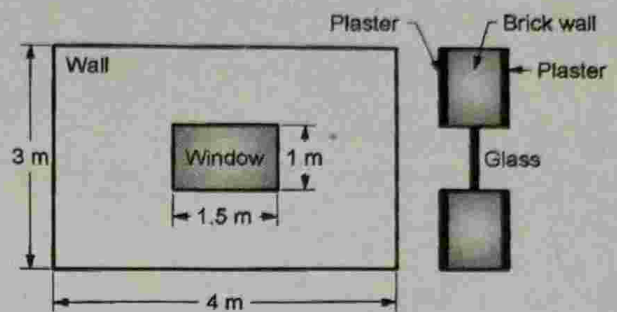
$$\text{Heat flow rate, } Q = \frac{T_1 - T_2}{\sum R} = \frac{1000 - 200}{0.01064}$$

$$= 75188 \text{ W or } 75.188 \text{ kW} \quad \dots \text{Ans.}$$

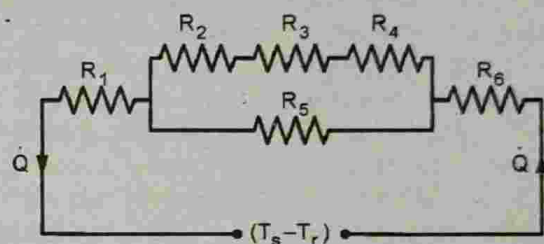
Ex. 2.12.9 : A room wall 4 m × 3 m of thickness 30 cm is made of brick of thermal conductivity 1.34 W/mK. It is covered on both sides with 20 mm thick plaster of thermal conductivity 0.8 W/mK. The wall has a window of size

(1.5 m × 1 m) having glass of thermal conductivity 0.75 W/mK and thickness of 12 mm. The inside temperature is maintained at 20°C and the surrounding temperature is 40°C. Find the heat gain by the room from the surroundings. Take heat transfer coefficient of air, $h = 15 \text{ W/m}^2 \text{K}$ for both in the room and surroundings.

Soln. : Refer Fig. P. 2.12.9.



(a)



(b)

Fig. P. 2.12.9

$$\text{Area of wall, } A_w = 4 \times 3 = 12 \text{ m}^2$$

$$\text{Area of window, } A_{wind} = 1.5 \times 1 = 1.5 \text{ m}^2$$

$$\text{Area of plaster, } A_p = A_w - A_{wind} = 12 - 1.5 = 10.5 \text{ m}^2$$

Convective resistance,

$$R_1 = R_6 = \frac{1}{h \cdot A_w} = \frac{1}{15 \times 12} = 0.00556 \text{ K/W}$$

Resistance for plaster,

$$R_2 = R_4 = \frac{x_p}{k_p \cdot A_p} = \frac{(20 \times 10^{-3})}{0.8 \times 10.5} = 0.00238 \text{ K/W}$$

Resistance for brick,

$$R_3 = \frac{x_w}{k_w A_p} = \frac{(30 \times 10^{-2})}{1.34 \times 10.5} = 0.02132 \text{ K/W}$$

Resistance for window glass,

$$R_5 = \frac{x_g}{k_g \cdot A_{wind}} = \frac{12 \times 10^{-3}}{0.75 \times 1.5} = 0.01067 \text{ K/W}$$



Equivalent resistance of the resistances in parallel,

$$\frac{1}{R_{eq}} = \frac{1}{R_2 + R_3 + R_4} + \frac{1}{R_5}$$

$$\frac{1}{R_{eq}} = \frac{1}{0.00238 + 0.02132 + 0.00238} + \frac{1}{0.01067}$$

$$R_{eq} = 0.00757 \text{ K/W}$$

∴ Combined resistance, $\Sigma R = R_1 + R_{eq} + R_6$

$$= 0.00556 + 0.00757 + 0.00556 = 0.01869 \text{ K/W}$$

Rate of heat gained by the room,

$$Q = \frac{\text{Surrounding temperature, } T_s - \text{Room Temperature, } T_r}{\Sigma R}$$

$$Q = \frac{40 - 20}{0.01869} = 1070.1 \text{ W} \quad \dots \text{Ans.}$$

Ex. 2.12.10 : A furnace wall is made up to 2 cm thick M.S. plate ($k = 55 \text{ W/mK}$) followed by asbestos insulation of 8 cm thickness ($k = 0.04 \text{ W/mK}$). Plate and the asbestos insulation are joined together by ten M.S. bolts ($k = 55 \text{ W/mK}$) of 10 mm diameter. Inside temperature of hot gases in the furnace is 250°C and outside air temperature is 30°C having the respective convective heat transfer coefficients as $30 \text{ W/m}^2\text{K}$ and $20 \text{ W/m}^2\text{K}$. Given : Area for both the furnace wall and insulating material as 3 m^2 excluding the area of bolt. Find the heat transfer rate.

Soln. : Refer Fig. P. 2.12.10.

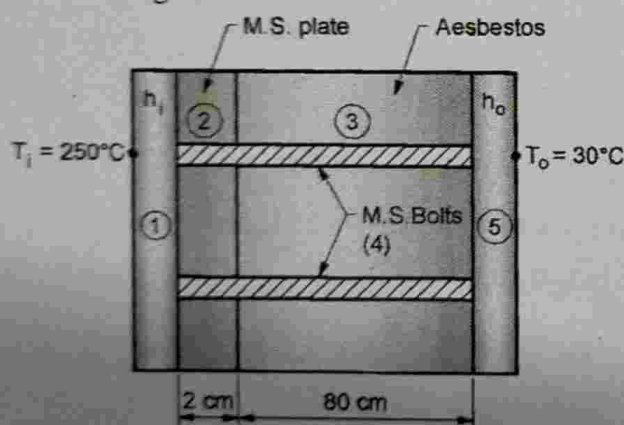


Fig. P. 2.12.10

$$k_2 (\text{M.S. plate}) = 55 \text{ W/mK} = k_4$$

$$k_3 (\text{Asbestos}) = 0.04 \text{ W/mK}$$

$$h_i = 30 \text{ W/m}^2\text{K}$$

$$h_o = 20 \text{ W/m}^2\text{K}$$

$$\text{Area of M. S. Plate / Asbestos} = A_2 = A_3 = 3 \text{ m}^2$$

$$\text{Area of ten bolts, } A_b = 10 \times \frac{\pi}{4} d_b^2$$

$$= 10 \times \frac{\pi}{4} \times \left(\frac{10}{1000}\right)^2$$

$$\text{Total area of (Wall + bolt) } A = 3 + 7.85 \times 10^{-4} \text{ m}^2$$

Resistance to inside convection

$$R_1 = \frac{1}{h_i A} = \frac{1}{30 \times (3 + 7.85 \times 10^{-4})} = 0.01111 \text{ K/W}$$

Resistance to outside convection,

$$R_5 = \frac{1}{h_o A} = \frac{1}{20 \times (3 + 7.85 \times 10^{-4})} = 0.01666 \text{ K/W}$$

Thermal resistance due to M.S. plate,

$$R_2 = \frac{x}{k_2 A_2} = \frac{2 \times 10^{-2}}{55 \times 3} = 0.00012 \text{ K/W}$$

Thermal resistance due to insulation,

$$R_3 = \frac{x}{k_3 A_3} = \frac{8 \times 10^{-2}}{0.04 \times 3} = 0.66667 \text{ K/W}$$

Thermal resistance due to bolts,

$$R_4 = \frac{x}{k_4 A_4} = \frac{(2 + 8) \times 10^{-2}}{55 \times (7.85 \times 10^{-4})} = 2.3161 \text{ K/W}$$

Equivalent resistance due to resistance in parallel we get,

$$\frac{1}{R_{eq}} = \frac{1}{(R_2 + R_3)} + \frac{1}{R_4}$$

$$\frac{1}{R_{eq}} = \frac{1}{0.01666 + 0.66667} + \frac{1}{2.3161}$$

$$R_{eq} = 0.52765 \text{ K/W}$$

Combined resistance,

$$\Sigma R = R_1 + R_{eq} + R_5$$

$$= 0.01111 + 0.52765 + 0.01666 = 0.55542 \text{ K/W}$$

Heat flow rate through the system,

$$Q = \frac{(T_1 - T_0)}{\Sigma R}$$

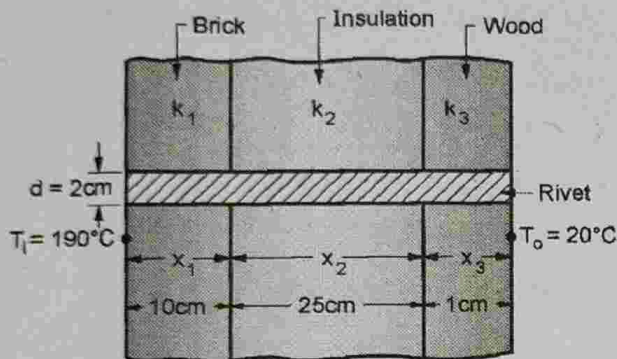
$$= \frac{(250 - 30)}{0.55542} = 396.1 \text{ W} \quad \dots \text{Ans.}$$

Ex. 2.12.11 : A composite insulating wall consists of three layers. These are held together by 2 cm diameter aluminium rivet per 0.1 m^2 of surface. The layers consists of 10 cm thick brick with hot surface at 190°C , 1 cm thick wood with cold surface at 20°C . These two layers are interposed by a third layer of insulating material 25 cm thick. Assuming one dimensional heat flow, calculate the percentage increase in heat transfer rate due to rivets. The conductivity of materials are :

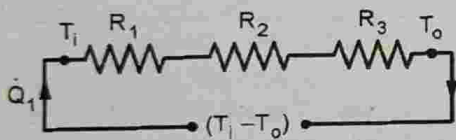
$$k_{\text{brick}} = 1 \text{ W/mK}, \quad k_{\text{insulation}} = 0.2 \text{ W/mK}, \\ k_{\text{Al}} = 220 \text{ W/mK}, \quad k_{\text{wood}} = 0.2 \text{ W/mK}.$$

Soln. :

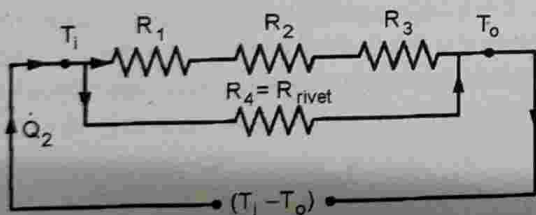
Refer Fig. P. 2.12.11.



(a) Heat transfer system



(b) Electrical system without rivet



(c) Electrical system with rivet

Fig. P. 2.12.11

Let surface area of composite wall,

$$A = 0.1 \text{ m}^2$$

Surface area of rivet,

$$A_r = \frac{\pi}{4} d^2 = \frac{\pi}{4} \left(\frac{2}{100} \right)^2 = 3.14 \times 10^{-4} \text{ m}^2$$

Resistance in the circuit are :

$$R_1 = \frac{x_1}{k_1 A} = \frac{(10/100)}{1 \times 0.1} = 1 \text{ K/W}$$

$$R_2 = \frac{x_2}{k_2 A} = \frac{(25/100)}{0.2 \times 0.1} = 12.5 \text{ K/W}$$

$$R_3 = \frac{x_3}{k_3 A} = \frac{(1/100)}{0.2 \times 0.1} = 0.5 \text{ K/W}$$

$$R_4 = R_{\text{rivet}} = \frac{x_4}{k_4 \cdot A_r} = \frac{(10 + 25 + 1)/100}{220 \times (3.14 \times 10^{-4})} \\ = 5.21 \text{ K/W}$$

(i) Heat transfer rate, Q_1 without rivet
[Fig. P. 2.12.11(a)]

$$Q_1 = \frac{(T_i - T_o)}{R_1 + R_2 + R_3} \\ = \frac{(190 - 20)}{1 + 12.5 + 0.5} = 12.143 \text{ W} \quad \dots \text{Ans.}$$

(ii) Heat transfer rate, Q_2 with rivet [Fig. P. 2.12.11(b)]

$$\frac{1}{\Sigma R} = \frac{1}{R_1 + R_2 + R_3} + \frac{1}{R_4} \\ = \frac{1}{1 + 12.5 + 0.5} + \frac{1}{5.21}$$

$$\Sigma R = 3.797 \text{ K/W}$$

$$\therefore Q_2 = \frac{(T_i - T_o)}{\Sigma R} \\ = \frac{(190 - 20)}{3.797} = 44.77 \text{ W} \quad \dots \text{Ans.}$$

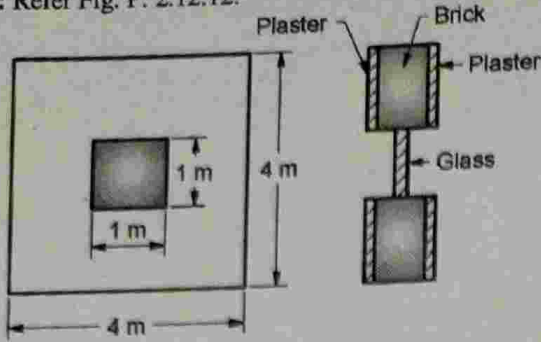
(iii) Percentage increase in heat transfer rate due to rivets,

$$= \left(\frac{Q_2 - Q_1}{Q_1} \right) \times 100 = \left(\frac{44.77 - 12.143}{12.143} \right) \times 100 \\ = 268.7 \% \quad \dots \text{Ans.}$$

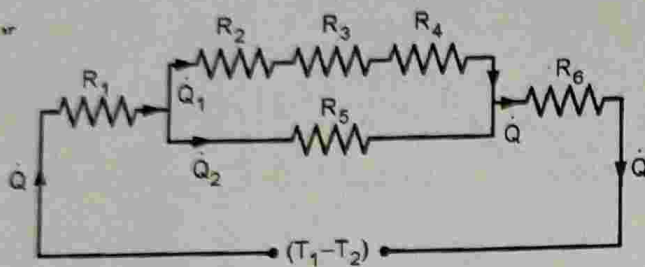
Ex. 2.12.12 : A $4 \text{ m} \times 4 \text{ m}$ wall consists of 15 cm thick layer of bricks with $k = 0.72 \text{ W/mK}$. It is covered with 2 cm thick plaster layer on both sides with $k = 0.1 \text{ W/mK}$. A through cavity of size $1 \text{ m} \times 1 \text{ m}$ is created in the middle of the wall and is completely fitted with a 1 cm thick glass window with $k = 0.78 \text{ W/mK}$. The assembly is exposed to air at 45°C on one side and air at 20°C on the other side. Heat transfer coefficient between plastered surface and air is $10 \text{ W/m}^2\text{K}$. Heat transfer coefficient between glass and air = $10 \text{ W/m}^2\text{K}$. Assume that, temperature of plastered surface = temperature of glass surface on each side. For one dimensional axial heat flow across the wall,

- Draw analogous electrical circuit for the arrangement.
- Calculate overall heat transfer coefficient.
- Calculate rate of heat flow.

Soln. : Refer Fig. P. 2.12.12.



(a) Heat transfer system



(b) Analogous electrical system

Fig. P. 2.12.12

$T_1 = 45^\circ\text{C}$, $T_2 = 20^\circ\text{C}$, $k_3 = k_{\text{brick}} = 0.72 \text{ W/mK}$;
 $x_3 = 15 \text{ cm} = 0.15 \text{ m}$, $k_{\text{plaster}} = k_2 = k_4 = 0.1 \text{ W/mK}$
 and

$x_2 = x_4 = 2 \text{ cm} = 0.02 \text{ m}$, $k_{\text{glass}} = k_5 = 0.78 \text{ W/mK}$
 and $x_5 = 1 \text{ cm} = 0.01 \text{ m}$, $h_1 = 10 \text{ W/m}^2\text{K} = h_6$

Area of wall, $A_w = 4 \times 4 = 16 \text{ m}^2$,

Area of glass, $A_g = 1 \times 1 = 1 \text{ m}^2$

$\therefore$ Area of brick and plaster, $A = A_w - A_g = 16 - 1 = 15 \text{ m}^2$

(i) Analogous electrical circuit is shown in Fig. P. 2.12.12(b) ...Ans.

(ii) Heat transfer rate, Q

Resistance due to convective heat transfer on both sides

$$R_1 = R_6 = \frac{1}{h \cdot A_w} = \frac{1}{10 \times 16} = 6.25 \times 10^{-3} \text{ K/W}$$

Resistance due to plaster,

$$R_2 = R_4 = \frac{x_2}{k_2 \cdot A} = \frac{0.02}{0.1 \times 15} = 13.33 \times 10^{-3} \text{ K/W}$$

Resistance due to brick,

$$R_3 = \frac{x_3}{k_3 \cdot A} = \frac{0.15}{0.72 \times 15} = 13.88 \times 10^{-3} \text{ K/W}$$

Resistance due to glass,

$$R_5 = \frac{x_5}{k_5 \cdot A_g} = \frac{0.01}{0.78 \times 1} = 12.82 \times 10^{-3}$$

Equivalent resistance of the loop (due to resistances in parallel),

$$\frac{1}{R_{\text{loop}}} = \frac{1}{R_2 + R_3 + R_4} + \frac{1}{R_5}$$

$$\frac{1}{R_{\text{loop}}} = \frac{1}{(13.33 \times 10^{-3} + 13.88 \times 10^{-3} + 13.33 \times 10^{-3})} + \frac{1}{12.82 \times 10^{-3}}$$

$$R_{\text{loop}} = 9.745 \times 10^{-3} \text{ K/W}$$

$$\Sigma R = R_1 + R_{\text{loop}} + R_6$$

$$= 6.25 \times 10^{-3} + 9.745 \times 10^{-3} + 6.25 \times 10^{-3}$$

$$= 22.245 \times 10^{-3} \text{ K/W}$$

$$Q = \frac{(T_1 - T_2)}{\Sigma R} = \frac{(45 - 20)}{22.245 \times 10^{-3}}$$

$$= 1123.84 \text{ W/m}^2$$

...Ans.

(iii) Overall heat transfer, coefficient, U

$$U = \frac{1}{\Sigma R} = \frac{1}{22.245 \times 10^{-3}}$$

$$= 44.954 \text{ W/m}^2\text{K}$$

...Ans.

Ex. 2.12.13 : An ice box (20 cm $\times$ 20 cm $\times$ 10 cm height) is filled with 3 kg of ice at 0°C . All vertical sides and bottom of the box are well insulated. The top cover of the box is 1 cm thick and is made up of a material with $k = 0.33 \text{ W/mK}$. This cover is exposed to ambient air at 25°C with $h = 10 \text{ W/m}^2\text{K}$. Heat transfer coefficient between inner surface of cover and air inside the box is $h = 8 \text{ W/m}^2\text{K}$. Calculate the time required to melt the ice in the box completely. Assume, latent heat of ice = 330 kJ/kg .

Soln. :

Refer Fig. P. 2.12.13. Heat will flow from outside to inside due to which ice melts. Heat flow is in vertical direction since all vertical sides and bottom is insulated.

$$h_o = 10 \text{ W/m}^2\text{K}, k = 0.33 \text{ W/mK}, h_i = 8 \text{ W/mK}.$$

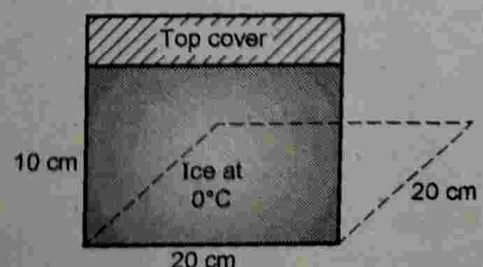


Fig. P. 2.12.13(Contd.)

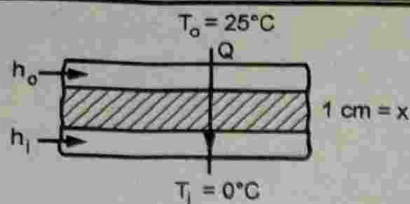


Fig. P. 2.12.13

Surface area of cover,

$$A = 20 \text{ cm} \times 20 \text{ cm} = \frac{20}{100} \times \frac{20}{100} = 0.04 \text{ m}^2$$

Combined resistance,

$$\Sigma R = \frac{1}{h_o A} + \frac{x}{k \cdot A} + \frac{1}{h_i A} = \frac{1}{A} \left[\frac{1}{h_o} + \frac{x}{k} + \frac{1}{h_i} \right]$$

$$\Sigma R = \frac{1}{0.04} \left[\frac{1}{10} + \frac{(1/100)}{0.33} + \frac{1}{8} \right] = 6.3826 \text{ K/W}$$

$$\dot{Q} = \frac{(T_o - T_i)}{\Sigma R} = \frac{(25 - 0)}{6.3826} = 3.917 \text{ W}$$

Energy required to melt $m_i = 3 \text{ kg}$ of ice will be,

$$Q = m_i \times \text{Latent heat of ice} = 3 \times 330 = 990 \text{ kJ}$$

$\therefore$ Time required for cooling,

$$t = \frac{Q}{\dot{Q}} = \frac{990 \times 10^3}{3.917} = 252.74 \times 10^3 \text{ s}$$

$$= \frac{252.74 \times 10^3}{3600} \text{ hrs} = 70.207 \text{ hrs} \quad \dots \text{Ans.}$$

Ex. 2.12.14 : A wall of size $3 \text{ m} \times 2 \text{ m}$ consists of 2 cm thick steel plate ($k = 15 \text{ W/mK}$) followed by 50 cm thick asbestos insulation ($k = 0.2 \text{ W/mK}$) and 4 cm thick layer of wood ($k = 0.3 \text{ W/mK}$). These three layers are held together with the help of mild steel bolts ($k = 40 \text{ W/mK}$), each of 12 mm diameter. Find the rate of heat flow through this composite wall when the temperature of hot gases on the outside of steel plate is maintained at 500°C with convective heat transfer coefficient of $40 \text{ W/m}^2\text{K}$ and the wooden layer is exposed to ambient air at 20°C with convective heat transfer coefficient of $12 \text{ W/m}^2\text{K}$.

Soln. : Refer Fig. P. 2.12.14.

Given :

Wall surface area, $A = 3 \times 2 = 6 \text{ m}^2$;

steel plate, $k_1 = 15 \text{ W/mK}$

Ten bolts made of steel, $k = 40 \text{ W/mK}$

$h_g = 40 \text{ W/m}^2\text{K}$, $h_a = 12 \text{ W/m}^2\text{K}$

Asbestos insulation, $k_2 = 0.2 \text{ W/mK}$

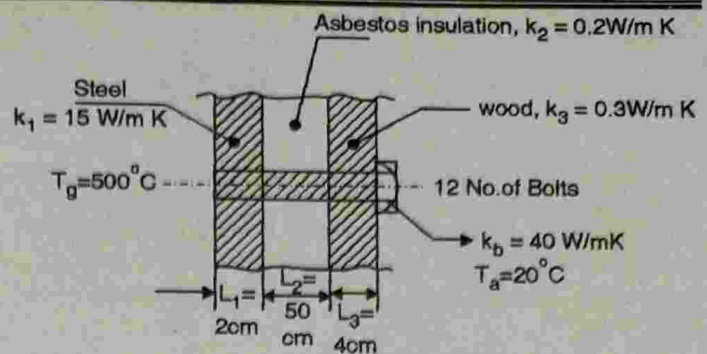


Fig. P. 2.12.14(a)

Given quantities are as shown in Fig. P. 2.12.14(a).

Cross sectional area of 12 bolts,

$$A_b = \frac{\pi}{4} (0.012)^2 \times 12 = 1.357 \times 10^{-3} \text{ m}^2$$

This cross sectional area of 12 number of bolts is very small compared with the surface area of the wall. Hence its effect is not considered.

Thermal resistances are as follows :

Gas, $R_1 = \frac{1}{h_g \cdot A} = \frac{1}{40 \times 6} = 4.167 \times 10^{-3} \text{ K/W}$

Steel plate, $R_1 = \frac{L_1}{k_1 \cdot A} = \frac{0.02}{15 \times 6} = 0.223 \times 10^{-3} \text{ K/W}$

Asbestos, $R_2 = \frac{L_2}{k_2 \cdot A} = \frac{0.5}{0.2 \times 6} = 416.67 \times 10^{-3} \text{ K/W}$

Wood, $R_3 = \frac{L_3}{k_3 \cdot A} = \frac{0.04}{0.3 \times 6} = 22.223 \times 10^{-3} \text{ K/W}$

Air, $R_0 = \frac{1}{h_a \cdot A} = \frac{1}{12 \times 6} = 13.888 \times 10^{-3} \text{ K/W}$

Bolts, $R_b = \frac{(L_1 + L_2 + L_3)}{k_b \cdot A_b}$
 $= \frac{(0.02 + 0.5 + 0.04)}{40 \times (1.357 \times 10^{-3})} = 10.316 \text{ K/W}$

Equivalent resistance of the resistances in parallel,

$$\frac{1}{R_{eq}} = \frac{1}{R_1 + R_2 + R_3} + \frac{1}{R_b}$$

$$\frac{1}{R_{eq}} = \frac{1}{(0.223 \times 10^{-3} + 416.67 \times 10^{-3} + 22.223 \times 10^{-3})} + \frac{1}{10.316}$$

$$R_{eq} = 0.4212 \text{ K/W}$$

$$\Sigma R = R_1 + R_{eq} + R_0$$

$$= 4.167 \times 10^{-3} + 0.4212 + 13.888 \times 10^{-3}$$

$$= 439.255 \times 10^{-3} \text{ K/W}$$

Heat transfer rate,

$$Q = \frac{T_s - T_a}{\sum R} = \frac{(500 - 20)}{439.255 \times 10^{-3}}$$

$$= 1092.8 \text{ W} \quad \dots \text{Ans.}$$

Ex. 2.12.15 : A homogenous wall of area 'A' and thickness 'L' has left and right surface temperatures at 0°C and 40°C respectively. Determine the temperature at the centre of the wall.

- How much material must be added and to which side of wall if the temperature at the centre is raised by 5°C ?
- How much material must be removed and from which side of the wall if the temperature at the centre line of the wall is to be lowered by 5°C ?

Soln. : Refer Fig. P. 2.12.15.

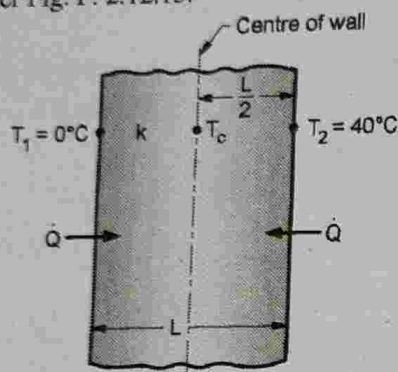


Fig. P. 2.12.15

Since, $Q = \frac{k A (T_2 - T_1)}{L} \quad \dots(i)$

For temperature at the middle of the wall, we can write,

$$Q = \frac{k \cdot A (T_2 - T_c)}{\left(\frac{L}{2}\right)} \quad \dots(ii)$$

On equating Equations (i) and (ii)

$$\frac{k A (T_2 - T_1)}{L} = \frac{k \cdot A \cdot (T_2 - T_c)}{\left(\frac{L}{2}\right)}$$

$$T_2 - T_1 = 2 (T_2 - T_c)$$

$$\text{i.e. } T_c = T_2 - \left(\frac{T_2 - T_1}{2}\right) = \frac{T_2 + T_1}{2}$$

$$= \frac{40 + 0}{2} = 20^\circ\text{C} \quad \dots \text{Ans.}$$

- When temperature at the centre of wall is raised by 5°C

$$T_c = 20 + 5 = 25^\circ\text{C}$$

Let x be the thickness of material added right side of the wall. Then modifying the Equation (i) we get,

$$Q = \frac{k A (T_2 - T_1)}{L + x} \quad \dots(iii)$$

For middle temperature

$$Q = \frac{k A (T_2 - T_c)}{\frac{L}{2} + x} \quad \dots(iv)$$

On equating Equations (iii) and (iv)

$$\frac{k A (T_2 - T_1)}{L + x} = \frac{k A (T_2 - T_c)}{\frac{L}{2} + x}$$

$$\frac{(40 - 0)}{L + x} = \frac{(40 - 25)}{\frac{L}{2} + x} \quad \dots(v)$$

On solving, $x = -\frac{L}{5}$

Negative sign indicates that the assumption of adding the wall on the right side is incorrect. **Thus the material of thickness $x = \frac{L}{5}$ must be added towards the left side of the wall so that $T_c = 25^\circ\text{C}$** ...Ans.

- When T_c is lowered by 5°C i.e. $T_c = 20 - 5 = 15^\circ\text{C}$

Let x be the material removed from right side. Again modifying the Equations (iii) and (iv) by equating,

$$\frac{k A (T_2 - T_1)}{L - x} = \frac{k A (T_2 - T_c)}{\frac{L}{2} - x}$$

$$\frac{40 - 0}{L - x} = \frac{40 - 15}{\frac{L}{2} - x}$$

$$20L - 40x = 25L - 25x$$

$$15x = -5L$$

$$x = -\frac{5L}{15} = -\frac{L}{3}$$

Negative sign shows that the material of $\frac{L}{3}$ thickness be removed from left side instead of from right side to get 15°C temperature at the middle of the wall. ...Ans.

Ex. 2.12.16 : A 2 kW heater element of area 0.04 m^2 is protected on the backside with insulation 50 mm thick of $k = 1.4 \text{ W/mK}$ and on the front side by a plate 10 mm thick with thermal conductivity of 45 W/mK . The backside is exposed to air at 5°C with convection coefficient of $10 \text{ W/m}^2\text{K}$ and the front is exposed to air at 15°C with convection coefficient including radiation of $250 \text{ W/m}^2\text{K}$. Determine the heater element temperature and heat flow into the room under steady state condition.

Soln. : Refer Fig. P. 2.12.16.

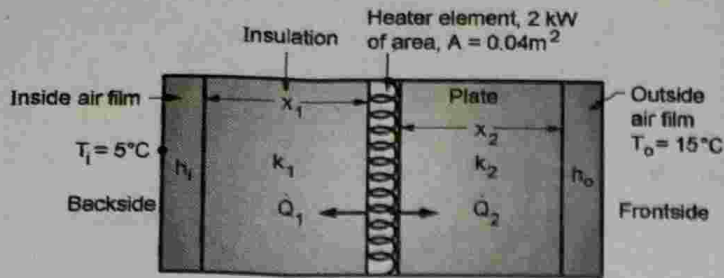


Fig. P. 2.12.16

Rating of heater, $Q = 2 \text{ kW} = 2000 \text{ W}$,

$$A = 0.04 \text{ m}^2$$

$$x_1 = 50 \text{ mm} = 0.05 \text{ m}, \quad k_1 = 1.4 \text{ W/mK},$$

$$h_1 = 10 \text{ W/m}^2\text{K}, \quad x_2 = 10 \text{ mm} = 0.01 \text{ m},$$

$$k_2 = 45 \text{ W/mK}, \quad h_0 = 250 \text{ W/m}^2\text{K}$$

Let T be the temperature of the heater element, for steady state conditions we have,

$$Q = \text{Heat flow rate backwards, } Q_1 \\ + \text{Heat flow rate front side, } Q_2$$

$$= \left[\frac{(T - T_1)}{\frac{x_1}{k_1 A} + \frac{1}{h_1 A}} \right] + \left[\frac{(T - T_0)}{\frac{x_2}{k_2 A} + \frac{1}{h_0 A}} \right]$$

$$2000 = \frac{(T - 5)}{\left[\frac{0.05}{1.4 \times 0.04} + \frac{1}{10 \times 0.04} \right]} + \frac{(T - 15)}{\left[\frac{0.01}{45 \times 0.04} + \frac{1}{250 \times 0.04} \right]}$$

$$\therefore 2000 = 0.2947 (T - 5) + 9.4737 (T - 15)$$

$$2000 = 0.2947 T - 1.4735 + 9.4737 T - 142.1055$$

$$2143.579 = 9.7684 T$$

$$T = 219.44^\circ\text{C} \quad \dots \text{Ans.}$$

Ex. 2.12.17 : A composite wall is shown in Fig. P. 2.12.17. The temperature at inside surface of the wall is 2000 K and the outside surrounding temperature is 300 K. The heat flux rate across the system is 500 W/m^2 . The convective heat transfer coefficient for air at the exterior surface is given by the equation, $h = 2.8 (\Delta T)^{0.3}$. Where, ΔT = Temperature difference between the outside surface and the surrounding temperature. Find the thickness x_2 and outside surface temperature.

Soln. :

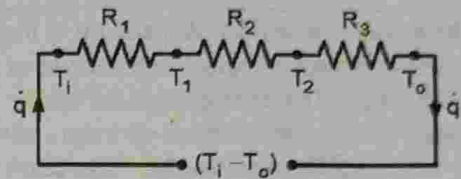
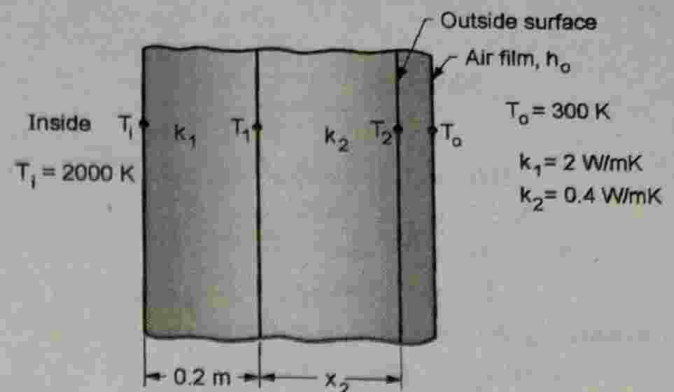


Fig. P. 2.12.17

$$\text{Heat flux rate, } q = \frac{Q}{A} = 500 \text{ W/m}^2;$$

$$h_0 = 2.8 (\Delta T)^{0.3} = 2.8 (T_2 - T_0)^{0.3}$$

$\therefore$ Heat flux rate remains constant, therefore,

$$q = \frac{T_1 - T_1}{R_1} = \frac{T_1 - T_2}{R_2} = \frac{T_2 - T_0}{R_3} \quad \dots (i)$$

The thermal resistances R_1 , R_2 and R_3 can be written as, ($A = 1 \text{ m}^2$)

$$R_1 = \frac{x_1}{k_1 A} = \frac{0.2}{2 \times 1} = 0.1 \text{ K/W} \quad \dots (ii)$$

$$R_2 = \frac{x_2}{k_2 A} = \frac{x_2}{0.4 \times 1} = 2.5 x_2 \text{ K/W} \quad \dots (iii)$$

$$R_3 = \frac{1}{h_0 A} = \frac{1}{2.8 (T_2 - T_0)^{0.3} \times 1} \text{ K/W} \quad \dots (iv)$$

From Equation (i),

$$q = \frac{T_1 - T_1}{R_1}$$

$$500 = \frac{2000 - T_1}{0.1}$$

$$T_1 = 1950 \text{ K}$$

$$\text{Also, } q = \frac{T_1 - T_2}{R_2}$$

$$\text{i.e. } 500 = \frac{1950 - T_2}{2.5 x_2}$$

$$1250 x_2 = 1950 - T_2 \quad \dots (v)$$



And, $q = \frac{T_2 - T_0}{R_3}$

i.e., $500 = \frac{T_2 - 300}{\frac{1}{2.8 (T_2 - 300)^{0.3}}}$

$$500 = (T_2 - 300) 2.8 (T_2 - 300)^{0.3}$$

$$178.57 = (T_2 - 300)^{1.3}$$

$$T_2 - 300 = 53.96$$

$$T_2 = 353.96$$

On substituting the value of T_2 is in Equation (v) we get,

$$1250 x_2 = 1950 - 353.96$$

$$x_2 = 1.277 \text{ m}$$

...Ans.

Ex. 2.12.18 : A wall of a furnace is made up of inside layer of silica brick 120 mm thick covered with a layer of magnesite brick 240 mm thick. The temperature at inside surface of silica brick wall and outside surface of magnesite brick wall are 725°C and 110°C respectively. The contact thermal resistance between the two walls at the interface is 0.0035°C/W per unit wall area. If thermal conductivities of silica and magnesite bricks are $1.7 \text{ W/m}^\circ\text{C}$ and $5.8 \text{ W/m}^\circ\text{C}$. Calculate.

(i) The rate of heat loss per unit area of walls.

(ii) The temperature drop at interface.

Soln. : Refer Fig. P. 2.12.18

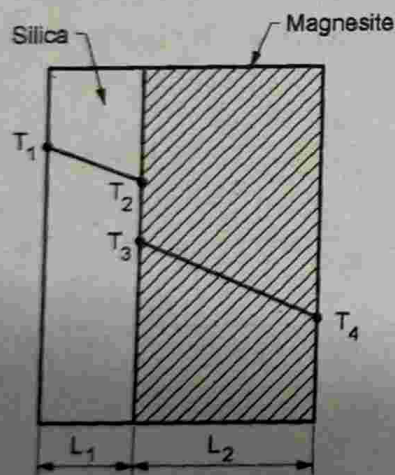


Fig. P. 2.12.18

Given data :

$$L_1 = 120 \text{ mm} = 0.12 \text{ m}$$

$$L_2 = 240 \text{ mm} = 0.24 \text{ m}$$

$$A_1 = \text{unit}$$

Contact resistance,

$$R_C = 0.0035^\circ\text{C/W}$$

$$k_1 = 1.7 \text{ W/m}^\circ\text{C}$$

$$k_2 = 5.8 \text{ W/m}^\circ\text{C}$$

$$T_1 = 725^\circ\text{C},$$

$$T_4 = 110^\circ\text{C}$$

(i) Rate of heat loss per unit area of walls, q

Thermal resistance of silica brick per m^2 surface area

$$R_1 = \frac{L_1}{k_1 A_1}$$

$$= \frac{0.12}{1.7 \times 1} = 0.0709 \text{ k/W}$$

$$R_2 = \frac{L_2}{k_2 A_2}$$

$$= \frac{0.24}{5.8 \times 1} = 0.04138 \text{ k/W}$$

Total resistance of system,

$$R_t = R_1 + R_C + R_2$$

$$= 0.070549 + 0.0035 + 0.04138$$

$$= 0.11547 \text{ k/W}$$

$$q = \frac{T_1 - T_4}{R_t} = \frac{725 - 110}{0.11547}$$

$$q = 5326 \text{ W/m}^2$$

...Ans.

(ii) Temperature drop at surface, $(T_2 - T_3)$

$$5326 = \frac{725 - T_2}{0.07059} = \frac{725 - T_3}{0.07059 + 0.0035}$$

$$q = \frac{T_1 - T_2}{R_1} = \frac{T_1 - T_3}{R_1 + R_C}$$

$$T_2 = 349.04^\circ\text{C}$$

$$T_3 = 330.40^\circ\text{C}$$

$$\therefore T_2 - T_3 = 349.04 - 330.4 = 18.64^\circ\text{C}$$

...Ans.

Ex. 2.12.19 : A circular plate heater (diameter 20 cm) is inserted between two circular plates (diameter 20 cm), slab A is 3 cm thick ($k = 55 \text{ W/mK}$) and slab B is 1.5 cm thick ($k = 0.18 \text{ W/mK}$). The outside heat transfer coefficient on sides of A and B are 200 and $65 \text{ W/m}^2 \text{ K}$ respectively. The temperature of surrounding air is 30°C . If the rating of the heater is 2 kW, find :

(i) Maximum temperature in the system.

(ii) Outer surface temperature of two slabs.

(iii) Draw equivalent electrical circuit of the system

Soln. :

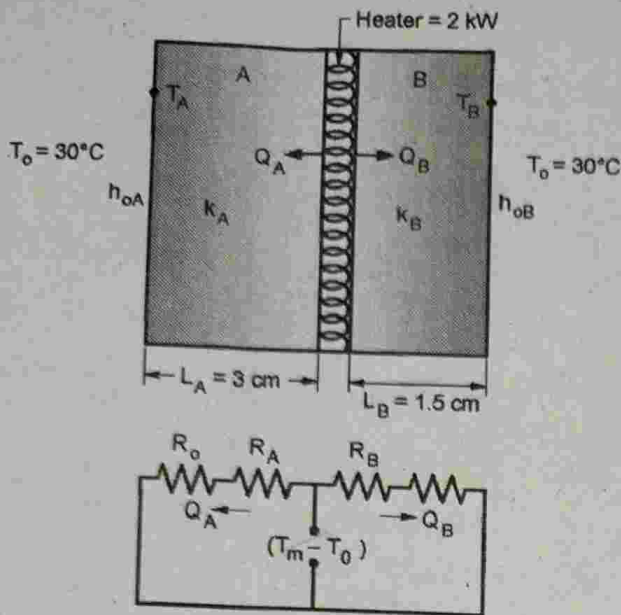


Fig. P. 2.12.19

Refer Fig. P. 2.12.19.

Diameter of plate heater = plates A and B = 20 cm = 0.2m

$$\therefore A = \frac{\pi}{4} \cdot d^2 = \frac{\pi}{4} (0.2)^2 = 0.03141 \text{ m}^2$$

$$h_{oA} = 200 \text{ W/m}^2 \text{ K}$$

$$h_{oB} = 65 \text{ W/m}^2 \text{ K}$$

$$L_A = 3 \text{ cm} = 0.03 \text{ m}$$

$$L_B = 1.5 \text{ cm} = 0.015 \text{ m}$$

$$k_A = 55 \text{ W/mK},$$

$$k_B = 0.18 \text{ W/mK}$$

Rating of heater, $Q = 2 \text{ kW} = 2000 \text{ W}$

$$T_{oA} = T_{oB} = T_o = 30^\circ\text{C}$$

(i) Maximum temperature, T_m

Total rate of heat produced by heater will be transferred through slabs A and B to surroundings. Let T_m be the maximum temperature at heater surface. Under steady flow conditions :

$$\begin{aligned} Q &= Q_A + Q_B \\ &= \frac{(T_m - T_o)}{\left(\frac{L_A}{k_A \cdot A} + \frac{1}{h_{oA} \cdot A}\right)} + \frac{(T_m - T_o)}{\left(\frac{L_B}{k_B \cdot A} + \frac{1}{h_{oB} \cdot A}\right)} \\ 2000 &= \frac{(T_m - 30) \times 0.03141}{\left(\frac{0.03}{55} + \frac{1}{200}\right)} + \frac{(T_m - 30) \times 0.03141}{\left(\frac{0.015}{0.18} + \frac{1}{65}\right)} \\ 2000 &= 0.03141 (T_m - 30) \times 190.46 \\ T_m &= 364.32^\circ\text{C} \quad \dots\text{Ans.} \end{aligned}$$

(ii) Outer surface temperatures of two slabs, T_A and T_B

$$\begin{aligned} Q_A &= \frac{k_A \cdot A \cdot (T_{\max} - T_A)}{L_A} \\ &= h_{oA} \cdot A \cdot (T_A - T_o) \\ \frac{k_A \cdot (T_{\max} - T_A)}{L_A} &= h_{oA} (T_A - T_o) \\ \frac{55(364.32 - T_A)}{0.03} &= 200 (T_A - 30) \\ 364.32 - T_A &= 0.10909 T_A - 3.2727 \\ T_A &= 331.44^\circ\text{C} \quad \dots\text{Ans.} \end{aligned}$$

Similarly,

$$\begin{aligned} k_B \frac{(T_{\max} - T_B)}{L_B} &= h_{oB} (T_B - T_o) \\ \frac{0.18(364.32 - T_B)}{0.015} &= 65(T_B - 30) \\ 364.32 - T_B &= 5.4167 T_B - 162.5 \\ T_B &= 82.1^\circ\text{C} \quad \dots\text{Ans.} \end{aligned}$$

(iii) Equivalent electrical circuit is shown in Fig. P. 2.12.19.

Ex. 2.12.20 : The temperature of the inner side of furnace wall is 640°C and that of outer side is 240°C and it is exposed to atmosphere at 40°C . In order to reduce the heat loss from the furnace, its wall thickness is increased by 100%. Calculate the percentage decrease in the heat loss due to increase in wall thickness. Assume no change in properties except temperature.

Soln. : Refer Fig. P. 2.12.20.

$$T_1 = 640^\circ\text{C}, T_2 = 240^\circ\text{C}, T_\infty = 40^\circ\text{C}$$

Let the thickness of the wall = b

New thickness, $b_1 = 2 \times \text{original thickness} = 2b$

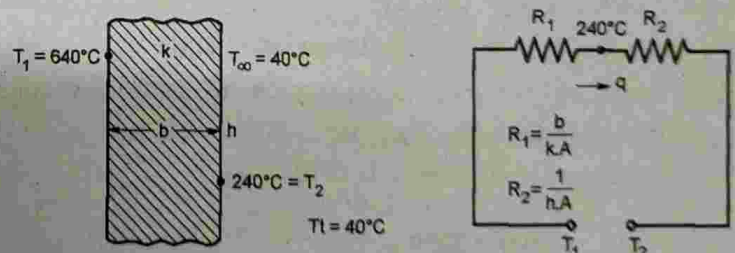


Fig. P. 2.12.20

Case I : Original wall

Initial rate of heat flow rate,

$$\begin{aligned} Q &= \frac{640 - 240}{\frac{b}{kA}} = \frac{240 - 40}{\frac{1}{hA}} \\ \therefore \frac{400k}{b} &= 200h; \quad h = \frac{2k}{b} \quad \dots(i) \end{aligned}$$

Heat transfer rate,

$$Q = \frac{T_1 - T_\infty}{R_1 + R_2} = \frac{T_1 - T_\infty}{\frac{b}{k \cdot A} + \frac{1}{h \cdot A}}$$

$$\therefore Q = \frac{A(640 - 40)}{\frac{b}{k} + \frac{1}{(2k/b)}} = \frac{A(600 \times 2k)}{3b} \quad \dots(ii)$$

Case II : When wall thickness is doubled, $b_1 = 2b$

$$Q_1 = \frac{T_1 - T_\infty}{R_1 + R_2} = \frac{T_1 - T_\infty}{\frac{b_1}{k \cdot A} + \frac{1}{h \cdot A}} = \frac{A(T_1 - T_\infty)}{\frac{2b}{k} + \frac{1}{(2k/b)}}$$

$$Q_1 = \frac{A(640 - 40) \times 2k}{5b} = \frac{A \times 600 \times 2k}{5b} \quad \dots(iii)$$

Percentage reduction in heat loss,

$$\begin{aligned} &= \left(\frac{Q - Q_1}{Q} \right) 100\% = \left(1 - \frac{Q_1}{Q} \right) 100 \\ &= \left[1 - \frac{A(600 \times 2k)/5b}{A(600 \times 2k)/3b} \right] \times 100 \\ &= \left(1 - \frac{3}{5} \right) 100 = 40\% \quad \dots Ans. \end{aligned}$$

Ex. 2.12.21 : A plane composite slab with unit cross sectional area is made up of material 'A' (thickness = 100 mm, $k_A = 60 \text{ W/mK}$) and material 'B' (thickness = 10 mm, $k_B = 2 \text{ W/mK}$) Thermal contact resistance at their interface is $0.003 \text{ m}^2 \text{ K/W}$. The temperature of open side of slab 'A' is 300°C and that of open side of slab 'B' is 50°C . Calculate,

- the rate of heat flow through the slab.
- temperatures on both sides of the interface.

Soln. : Refer Fig. P. 2.12.21.

Let t_2 and t_3 represent the temperature at the theoretical plane interface obtained consideration of heat flow in materials on either side. Then for a given wall area A the thermal constant resistance R_c is defined as,

$$R_c = 0.003 \text{ K/W} = \frac{T_2 - T_3}{Q}$$

Let, $A = 1 \text{ m}^2$

$$L_A = 100 \text{ mm} = 0.1 \text{ m}, k_A = 60 \text{ W/mK}$$

$$L_B = 10 \text{ mm} = 0.01 \text{ m}, k_B = 2 \text{ W/mK}$$

$$T_1 = 300^\circ\text{C}, T_4 = 50^\circ\text{C}$$

$$R_A = \frac{L_A}{k_A \cdot A} = \frac{0.1}{60} = 1.667 \times 10^{-3} \text{ K/W}$$

$$R_B = \frac{L_B}{k_B \cdot A} = \frac{0.01}{2} = 5 \times 10^{-3} \text{ K/W}$$

$$\begin{aligned} \text{Total resistance, } \Sigma R &= R_A + R_c + R_B \\ &= 1.667 \times 10^{-3} + 0.003 + 5 \times 10^{-3} \\ &= 9.667 \times 10^{-3} \text{ W/K} \end{aligned}$$

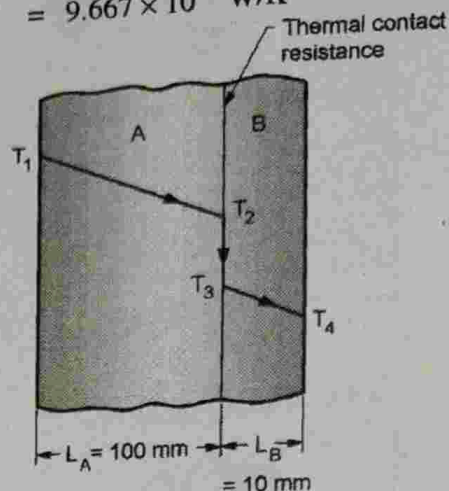


Fig. P. 2.12.21

- Rate of heat flow through slab, Q

$$\begin{aligned} Q &= \frac{T_1 - T_4}{\Sigma R} = \frac{(300 - 50)}{9.667 \times 10^{-3}} \\ &= 25.861 \times 10^3 \text{ W} \quad \dots Ans. \end{aligned}$$

- Temperature, T_2 and T_3 on Left and Right side of Interface

$$Q = \frac{T_1 - T_2}{R_A};$$

$$T_2 = T_1 - Q \cdot R_A$$

$$\begin{aligned} T_2 &= 300 - (25.861 \times 10^3) 1.667 \times 10^{-3} \\ &= 256.89^\circ\text{C} \quad \dots Ans. \end{aligned}$$

$$Q = \frac{T_2 - T_3}{R_c}$$

$$25.861 \times 10^{-3} = \frac{256.89 - T_3}{0.003}$$

$$T_3 = 179.307^\circ\text{C} \quad \dots Ans.$$

Ex. 2.12.22 : A layer of 5 cm refractory brick ($k = 2 \text{ W/m K}$) is to be placed between two 4 mm thick steel plates ($k = 40 \text{ W/m. K}$). Both faces of brick adjacent to the plates have rough solid to solid contact over 20% of the area, where the average height of asperities are 1 mm. The outer surface temperature of steel plates are 400°C and 100°C respectively.

- Find the rate of heat loss per unit area assuming that the cavity area is filled with air ($k = 0.02 \text{ W/m. K}$.)
- Find the rate of heat flow if the faces of the brick are smooth and have solid to solid contact perfectly over entire area.

Soln. : Refer Fig. P. 2.12.22(a) and equivalent electrical circuit is shown in Fig. P. 2.12.22(b).

Brick : $b_2 = 5 \text{ cm} = 0.05 \text{ m}$, $k_2 = 2 \text{ W/mK}$

Steel plate : $b_1 = 4 \text{ mm} = 0.004 \text{ m}$, $k_1 = 40 \text{ W/mK}$

Cavity filled with air : $k_3 = 0.02 \text{ W/mK}$,

$$b_3 = 1 \text{ mm} = 0.001 \text{ m}$$

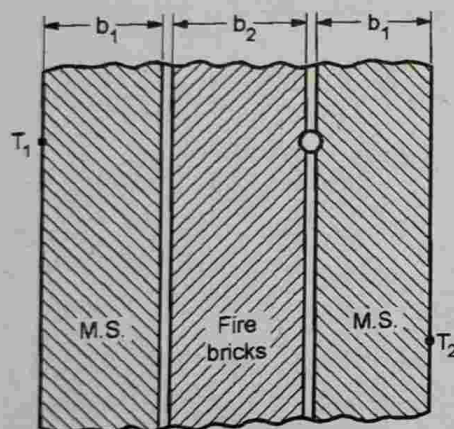
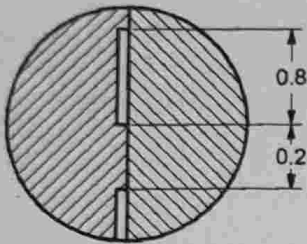
$$\frac{\text{Plate area, } A_s}{\text{Brick area, } A} = 20\% = 0.2, \quad T_1 = 400^\circ\text{C}, \quad T_2 = 100^\circ\text{C}$$

Consider unit area of steel plate i.e. $A_1 = 1 \text{ m}^2$.

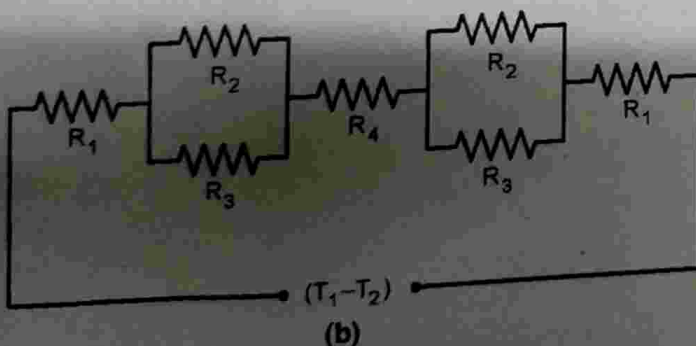
$$\text{Area of brick surface, } A_2 = 0.2 A_1 = 0.2 \times 1 = 0.2 \text{ m}^2$$

Area of air gap,

$$A_3 = A_1 - 0.2 A_1 = 0.8 A_1 = 0.8 \times 1 = 0.8 \text{ m}^2.$$



(a)



(b)

Fig. P. 2.12.22

Thermal resistances are as follows :

$$\text{Steel plates, } R_1 = \frac{b_1}{k_1 \cdot A} = \frac{0.004}{40 \times 1} = 0.0001 \text{ K/W}$$

$$\text{Air gap, } R_3 = \frac{b_3}{k_3 \cdot A_3} = \frac{0.001}{0.02 \times 0.8} = 0.0625 \text{ K/W}$$

$$\text{Fire bricks, } R_2 = \frac{b_2}{k_2 \cdot A_2} = \frac{0.05}{2 \times 0.2} = 0.0025 \text{ K/W}$$

Thickness of layer of fire bricks

As the roughness of brick surface is relatively larger than the steel surface, we can assume that the aspirates are inside the brick only. Hence the solid layer of brick has thickness,

$$b_2 = (\text{Thickness of slab} - \text{Thickness of aspirates}) \\ = 50 - 2 = 48 \text{ mm}$$

$$R_4 = \frac{b_2}{k_2 \cdot A} = \frac{0.048}{2 \times 1} = 0.024 \text{ K/W}$$

Contact resistance,

$$\frac{1}{R_c} = \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{0.0625} + \frac{1}{0.0025}$$

$$R_c = 0.024 \text{ K/W}$$

Equivalent resistance to the system,

$$\begin{aligned} \Sigma R &= R_1 + R_c + R_4 + R_c + R_1 \\ &= 0.0001 + 0.0024 + 0.0240 + 0.0024 + 0.0001 \\ &= 0.029 \text{ K/W} \end{aligned}$$

(i) Heat flow rate, Q

$$Q = \frac{(T_1 - T_2)}{\Sigma R} = \frac{400 - 100}{0.029} = 10344.8 \text{ W} \quad \dots \text{Ans.}$$

(ii) Rate of heat flow for solid contact on steel plate over the entire area, Q_1

$$\text{Steel, } R_1 = \frac{b_1}{k_1 \cdot A} = \frac{0.004}{40 \times 1} = 0.0001 \text{ K/W}$$

$$\text{Brick, } R_b = \frac{b_2}{k_2 \cdot A} = \frac{0.05}{2 \times 1} = 0.0250 \text{ K/W}$$

$$\Sigma R = R_1 + R_b = 0.0001 + 0.0250 = 0.0251 \text{ K/W}$$

$$\begin{aligned} Q_1 &= \frac{T_1 - T_2}{\Sigma R} \\ &= \frac{(400 - 100)}{0.0251} = 11952.2 \text{ W} \quad \dots \text{Ans.} \end{aligned}$$

Ex. 2.12.23 : A wall 30 cm thick of size 5 m × 3 m made of red bricks ($k = 0.35 \text{ W/m K}$). It is covered on both sides by the layers of plaster 2 cm thick $k = 0.6 \text{ W/m k}$. The wall has a window of size 1 m × 2 m. The window door is made of glass, 12 mm thick having thermal conductivity 1.2 W/mK. Estimate the rate of heat flow through the wall. Inner and outer surface temperatures are 10°C and 40°C, respectively. Take $h = 15 \text{ W/m}^2 \text{ K}$ on both sides of wall.

Soln. :

$$\text{Area of wall} = 5 \times 3 = 15 \text{ m}^2 = A_w$$

$$\text{Area of window} = 1 \times 2 = 2 \text{ m}^2 = A_{\text{wind}}$$

$$\text{Area of plaster} = A_w - A_{\text{wind}} = 13 \text{ m}^2$$

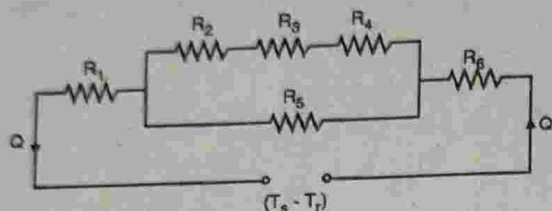
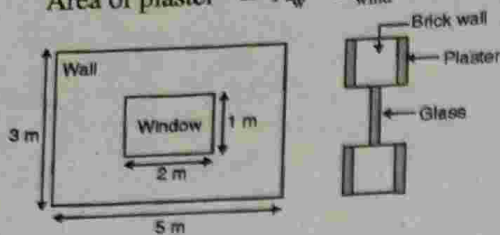


Fig. P. 2.12.23

Convective resistance

$$R_1 = R_6 = \frac{1}{h \cdot A_w} = \frac{1}{15 \times 15} = 0.0044 \text{ K/W}$$

Resistance for plaster

$$R_2 = R_4 = \frac{X_p}{k_p \cdot A_p} = \frac{2 \times 10^{-2}}{0.6 \times 13} = 0.00256 \text{ K/W}$$

Resistance for brick

$$R_3 = \frac{X_w}{k_w \cdot A_p} = \frac{30 \times 10^{-2}}{0.35 \times 13} = 0.06593 \text{ K/W}$$

Resistance for window glass

$$R_5 = \frac{X_g}{k_g \cdot A_{\text{wind}}} = \frac{12 \times 10^{-3}}{1.2 \times 2} = 0.005 \text{ K/W}$$

Equivalent resistance of the resistances in parallel

$$\frac{1}{R_{eq}} = \frac{1}{R_2 + R_3 + R_4} + \frac{1}{R_5}$$

$$\frac{1}{R_{eq}} = \frac{1}{0.00256 + 0.0659 + 0.00256} + \frac{1}{0.005}$$

$$\therefore \frac{1}{R_{eq}} = \frac{1}{0.07102} + \frac{1}{0.005}$$

$$\frac{1}{R_{eq}} = 14.08 + 200$$

$$\therefore R_{eq} = 4.67 \times 10^{-3}$$

$\therefore$ Combined resistance

$$\begin{aligned} \Sigma R &= R_1 + R_{eq} + R_6 = 0.0044 + 4.67 \times 10^{-3} + 0.0044 \\ &= 0.01347 \text{ k/W} \end{aligned}$$

Rate of heat gained by the room

$$Q = \frac{\text{Surrounding temperature } T_s - \text{Room temperature } T_r}{\Sigma R}$$

$$\therefore Q = \frac{40 - 10}{0.01347} = 2227 \text{ W}$$

...Ans.

Ex. 2.12.24 : The interior of a refrigerator has inside dimensions 60 cm x 45 cm base area and 120 cm high. The composite wall is made of two 3 mm mild steel sheets ($k = 145 \text{ kJ/m-hr-deg}$) with 6 cm of glass wool ($k = 0.188 \text{ kJ/m-hr-deg}$) insulation sandwiched between them. The average values of convective heat transfer coefficients at the interior and exterior wall are 40.8 and 52.3 $\text{kJ/m}^2\text{-hr-deg}$ respectively.

- Draw the thermal circuit.
- Calculate the individual resistance of this composite wall and the resistance at the surface, and the overall conductance.
- For the air temperature inside the refrigerator at 6.5°C and outside of 25°C, determine the rate at which heat must be removed from the refrigerator. Also, calculate the temperature on the outer surface of the metal sheet.

Soln. :

Given : $\Delta x_1 = \Delta x_3 = 3 \text{ mm} = 0.003 \text{ m}$; $\Delta x_2 = 0.06 \text{ m}$

$$k_2 = \frac{0.188 \times 1000}{3600} = 0.052 \text{ W/mK}$$

$$k_1 = k_3 = \frac{145 \times 1000}{3600} = 40.278 \text{ W/mK}$$

$$h_1 = \frac{40.8 \times 1000}{3600} = 11.33 \text{ W/m}^2\text{K}$$

$$h_o = \frac{52.3 \times 1000}{3600} = 14.53 \text{ W/m}^2\text{K}$$

Area transferring heat

$$\begin{aligned} A &= 0.60 \times 0.45 \times 2 + 0.60 \times 1.20 \times 2 + 0.45 \times 1.2 \times 2 \\ &= 0.54 + 1.44 + 1.08 = 3.06 \text{ m}^2 \end{aligned}$$

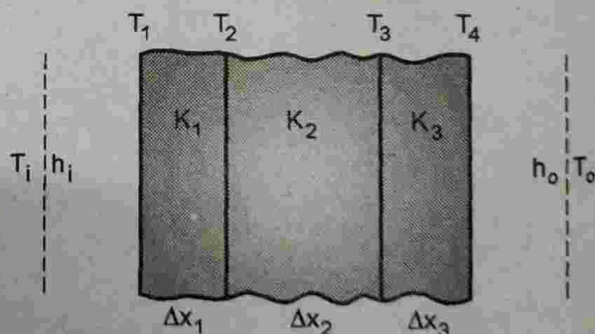
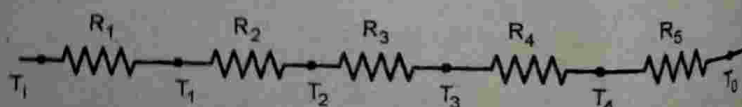


Fig. P. 2.12.24

(i) Thermal circuit



$$(ii) R_1 = \frac{1}{h_1 A} = \frac{1}{11.33 \times 3.06}$$

$$= 28.84 \times 10^{-3} \text{ K/W}$$

...Ans.

$$R_2 = \frac{\Delta x_1}{k_1 A} = \frac{0.003}{40.278 \times 3.06}$$

$$= 0.0024 \times 10^{-3} \text{ K/W} = R_4$$

...Ans.

$$R_3 = \frac{\Delta x_2}{k_2 A} = \frac{0.06}{0.052 \times 3.06}$$

$$= 377 \times 10^{-3} \text{ K/W}$$

...Ans.

$$R_5 = \frac{1}{h_o A} = \frac{1}{14.53 \times 3.06}$$

$$= 22.49 \times 10^{-3} \text{ K/W}$$

...Ans.

$$\text{Total } R = (28.84 + 0.0024 + 377 + 0.0024 + 22.49) \times 10^{-3} \text{ K/W} = 428.34 \times 10^{-3} \text{ K/W}$$

...Ans.

∴ Overall conductance

$$U = \frac{1}{R \times A} = \frac{1}{428.34 \times 10^{-3} \times 3.06}$$

$$= 0.763 \text{ W/m}^2\text{K}$$

...Ans.

$$(iii) \quad Q = \frac{\Delta T}{R} \frac{25 - 6.5}{428.34 \times 10^{-3}} = 43.19 \text{ W}$$

...Ans.

Also, to find T_a :

$$Q = \frac{T_0 - T_4}{R_5}$$

$$43.19 \times 22.49 \times 10^{-3} = 25 - T_4$$

$$T_4 = 24.03^\circ\text{C}$$

...Ans.

Ex. 2.12.25 : The inside temperature of furnace wall 200 mm thick is 1350°C and mean thermal conductivity of wall material is 1.35 W/mK . The heat transfer coefficient of outside surface is a function of temperature difference $h = 7.85 + 0.08 \Delta T$, where ΔT is the temperature difference between outside wall surface and surrounding. Determine the rate of heat transfer per unit area if surrounding temperature is 40°C .

Soln. :

$$h = 7.85 + 0.08 \Delta T$$

$$Q = \frac{T_1 - T_2}{\frac{\Delta x}{k}} = \frac{T_2 - T_\infty}{\frac{1}{h}}$$

$$= \frac{1350 - T_2}{\frac{0.2}{1.35}} = \frac{T_2 - 40}{\frac{1}{7.85 + 0.08(T_2 - 40)}}$$

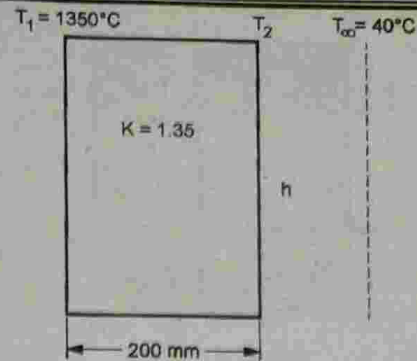


Fig. P. 2.12.25

$$6.75(1350 - T_2) = [7.85 + 0.08(T_2 - 40)](T_2 - 40)$$

$$9112.5 - 6.75 T_2 = [7.85 + 0.08 T_2 - 3.2](T_2 - 40)$$

$$= 7.85 T_2 + 0.08 T_2^2 - 3.2 T_2$$

$$- 314 - 3.2 T_2 + 128$$

$$0.08 T_2^2 + 8.2 T_2 - 9298.5 = 0$$

$$T_2^2 + 102.5 T_2 - 116231 = 0$$

By trial and error, $T_2 = 293.6^\circ\text{C}$.

∴ Heat transfer rate,

$$Q = \frac{1350 - 293.6}{\frac{0.2}{1.35}} = 7130.7 \text{ W/m}^2 \quad \dots\text{Ans.}$$

Ex. 2.12.26 : Find the heat flux across a composite slab of width 0.25 m and 0.15 m of conductivity 388 W/m-K and 250 W/m-K respectively when its one surface is at 150°C and the other surface is at 40°C . Also find the temperature at the joint of two materials of the slab.

Soln. :**Given :**

$$\Delta x_1 = 0.25 \text{ m}; \quad \Delta x_2 = 0.15 \text{ m}$$

$$T_1 = 150^\circ\text{C}; \quad k_1 = 388 \text{ W/m-K}$$

$$T_3 = 40^\circ\text{C}; \quad k_2 = 250 \text{ W/m-K}$$

(i) Heat flow per unit area

$$\text{Heat flux } q = \frac{T_1 - T_3}{\frac{\Delta x_1}{k_1} + \frac{\Delta x_2}{k_2}} = \frac{150 - 40}{\frac{0.25}{388} + \frac{0.15}{250}} = 88403 \text{ W/m}^2$$

...Ans.

(ii) Temperature at the joint of two material, T_2

$$q = \frac{T_1 - T_2}{\frac{\Delta x_1}{k_1}} \Rightarrow 88403 = \frac{150 - T_2}{\frac{0.25}{388}}$$

$$T_2 = 93^\circ\text{C}$$

...Ans.

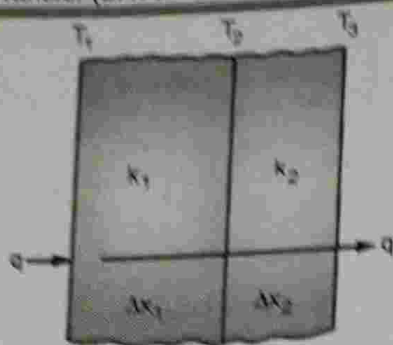


Fig. P. 2.12.26

Ex. 2.12.27 : The composite wall of a furnace is made up with 120 mm of fire clay [$k = 0.25 (1 + 0.0009 t)$ W/m°C] and 600 mm of red brick ($k = 0.8$ W/m°C). The inside surface temperature is 1250°C and the outside temperature is 40°C. Determine :

- The temperature at the layer interface and
- The heat loss for 1 m² of surface wall.

Soln. :

$$k_f = 0.25 (1 + 0.0009 T) \text{ W/m}^2\text{°C}$$

$$k_r = 0.8 \text{ W/m}^2\text{°C} ; T_1 = 1250^\circ\text{C}$$

$$T_3 = 40^\circ\text{C}$$

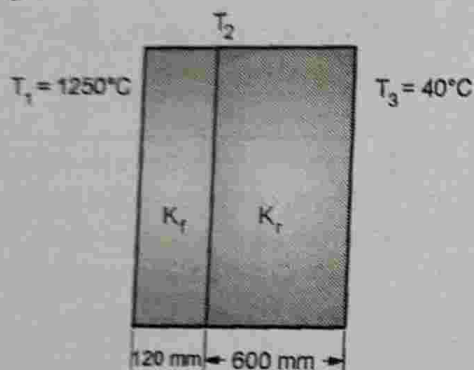


Fig. P. 2.12.27

Mean conductivity of fire clay,

$$\begin{aligned} k_{\text{mf}} &= 0.25 \left[1 + 0.0009 \left(\frac{1250 + T_2}{2} \right) \right] \\ &= 0.25 + 0.14 + 1.125 \times 10^{-4} T_2 \\ &= 0.39 + 1.125 \times 10^{-4} T_2 \end{aligned}$$

Heat flow rate,

$$\begin{aligned} Q &= \frac{1250 - 40}{\frac{0.12}{0.39 + 1.125 \times 10^{-4} T_2} + \frac{0.6}{0.8}} \\ &= \frac{1210}{\frac{1}{3.25 + 9.375 \times 10^{-4} T_2} + 0.75} \end{aligned}$$

Also,

$$Q = \frac{T_2 - 40}{0.8}$$

$$\therefore \frac{1210}{\frac{1}{3.25 + 9.375 \times 10^{-4} T_2} + 0.75} = \frac{T_2 - 40}{0.8}$$

Simplifying,

$$0.000703 T_2^2 + 2.505 T_2 - 3287.5 = 0$$

$$\text{or } T_2^2 + 3563.3 T_2 - 4676387 = 0$$

$$T_2 = \frac{-3563.3 \pm \sqrt{3563.3^2 - 4 \times 1 \times (-4676387)}}{2}$$

$$\text{Solving, } T_2 = 1020.3^\circ\text{C}$$

$$\begin{aligned} \therefore \text{Heat flow rate } Q &= \frac{T_2 - T_3}{\frac{0.6}{0.8}} \\ &= \frac{1020.3 - 40}{0.75} \\ &= 1307 \text{ W/m}^2 \end{aligned}$$

...Ans.

Ex. 2.12.28 : A steel pipe is insulated which carries hot liquid. Internal diameter of pipe is 10 cm of wall thickness 1 cm and thickness of insulation laid on pipe is 3 cm. Temperature of hot liquid is 85°C and that of surroundings is 25°C. Thermal conductivity of steel is 58 W/mK and that of insulation is 0.2 W/mK respectively. Heat transfer coefficient on inside and outside surface are 720 W/m²K and 9 W/m²K respectively. Find the heat loss per meter length of pipe.

Soln. :

$$\text{Given : } r_1 = \frac{10}{2} = 5 \text{ cm, } r_2 = 5 + 1 = 6 \text{ cm, } r_3 = 6 + 3 = 9 \text{ cm}$$

$$h_i = 720 \text{ W/m}^2\text{K, } k_1 = 58 \text{ W/mK,}$$

$$k_2 = 0.2 \text{ W/mK, } h_o = 9 \text{ W/m}^2\text{K,}$$

$$T_i = 85^\circ\text{C, } T_o = 25^\circ\text{C}$$

$$\text{and } L = 1 \text{ m.}$$

$$Q = \frac{T_i - T_o}{\frac{1}{h_i 2\pi r_1 L} + \frac{\log_e \frac{r_2}{r_1}}{2\pi k_1 L} + \frac{\log_e \frac{r_3}{r_2}}{2\pi k_2 L} + \frac{1}{h_o 2\pi r_3 L}}$$

$$\begin{aligned} \therefore \frac{Q}{L} &= \frac{2\pi (T_i - T_o)}{\frac{1}{h_i r_1} + \frac{\log_e \frac{r_2}{r_1}}{k_1} + \frac{\log_e \frac{r_3}{r_2}}{k_2} + \frac{1}{h_o r_3}} \\ &= \frac{2 \times 3.14 (85 - 25)}{\left(\frac{1}{720 \times 0.05} + \frac{\log_e \frac{6}{5}}{58} + \frac{\log_e \frac{9}{6}}{0.2} + \frac{1}{9 \times 0.09} \right)} \\ &= 114.63 \text{ W/m} \end{aligned}$$

...Ans.



Ex. 2.12.29 : A pipe of inner diameter 100 mm and outer diameter 120 mm carries steam at 110°C . The thermal conductivity of pipe material is $185 \text{ W/m}^\circ\text{C}$. The pipe is located in a room where the ambient temperature is 30°C and the convective heat transfer coefficient between pipe and air is $15 \text{ W/m}^2\text{C}$. Determine heat transfer rate per unit length of pipe. If the pipe is covered with 50 mm thick insulation of thermal conductivity $0.2 \text{ W/m}^\circ\text{C}$ find the heat transfer rate per unit length of pipe.

Soln. :

Refer Fig. P. 2.12.29.

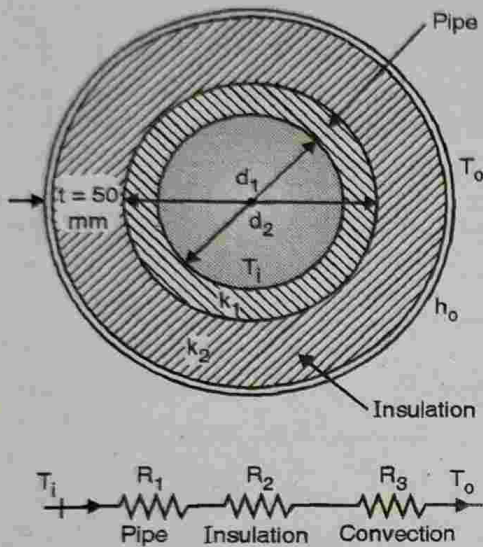


Fig. P. 2.12.29

Given :

Since h value on inner surface not given, inner surface temperature will have to be assumed that of steam.

$$d_1 = 100 \text{ mm}, \quad d_2 = 120 \text{ mm}$$

$$r_1 = \frac{100}{2} = 50 \text{ mm}, \quad r_2 = \frac{120}{2} = 60 \text{ mm}$$

$$k_1 = 185 \text{ W/m}^\circ\text{C} \text{ (pipe material)}$$

$$T_i = 110^\circ\text{C}, \quad T_o = 30^\circ\text{C},$$

$$h_o = 15 \text{ W/m}^2\text{C}, \quad L = 1 \text{ m}$$

Insulation, $t = 50 \text{ mm}$ thick, therefore,

$$d_3 = d_2 + 2t = 120 + 2 \times 50 = 220 \text{ mm}$$

$$r_3 = \frac{220}{2} = 110 \text{ mm}, \quad k_2 = 0.2 \text{ W/m}^\circ\text{C} \text{ (insulation)}$$

Consider unit length of pipe, $L = 1 \text{ m}$.

Thermal resistance of pipe,

$$R_1 = \frac{\log_e \frac{r_2}{r_1}}{2\pi k_1 L} = \frac{\log_e \left(\frac{60}{50} \right)}{2\pi \times 185 \times 1} = 1.569 \times 10^{-4}$$

$$R_2 = \frac{\log_e \frac{r_3}{r_2}}{2\pi k_2 L} = \frac{\log_e \left(\frac{110}{60} \right)}{2\pi \times 0.2} = 0.48235$$

$$R_3 = \frac{1}{h_o \cdot 2\pi r_3 L} = \frac{1}{15 \times 2\pi \times \frac{110}{1000} \times 1} = 0.09646$$

$$\therefore \text{Total resistance, } R = R_1 + R_2 + R_3$$

$$= 1.569 \times 10^{-4} + 0.48235 + 0.09646 = 0.57897$$

Heat transfer rate Q

$$Q = \frac{T_i - T_o}{R} = \frac{(110 - 30)}{0.57897}$$

$$= 138.18 \text{ W/m length of pipe}$$

...Ans.

Ex. 2.12.30 : A spherical tank, 1 m in diameter is maintained at a temperature of 120°C and exposed to a convection environment with $h = 25 \text{ W/m}^2\text{K}$ and temperature of ambient is 15°C . What thickness of urethane foam ($k = 20 \times 10^{-3} \text{ W/mK}$) should be added to ensure that the outer temperature of the insulation does not exceed 40°C ? What percentage reduction in heat loss results from installing this insulation?

Soln. : Refer Fig. P. 2.12.30

$$d_1 = 1 \text{ m}; \quad r_1 = 0.5 \text{ m}$$

$$T_1 = 120^\circ\text{C}; \quad h = 25 \text{ W/m}^2\text{K}$$

$$T_\infty = 15^\circ\text{C} \quad k = 20 \times 10^{-3} \text{ W/mK}$$

$$T_2 = 40^\circ\text{C}$$

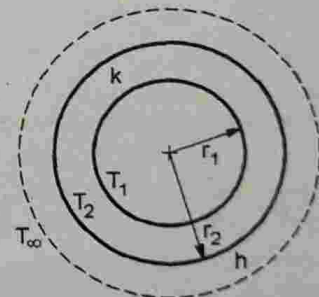


Fig. P. 2.12.30

(i) Heat transfer without insulation from tank, Q_1

$$Q_1 = \frac{T_1 - T_\infty}{\frac{1}{hA}} = \frac{(120 - 15)}{\left(\frac{1}{25 \times 4\pi \times 0.5^2} \right)}$$

$$= 8242.5 \text{ W}$$

...Ans.

(ii) Radius of insulation, r_2 when $T_2 = 40^\circ\text{C}$

$$Q_2 = \frac{T_2 - T_\infty}{\frac{1}{h \cdot 4\pi r_2^2}} = \frac{40 - 15}{\frac{1}{25 \times 4\pi \times r_2^2}} = 7850 r_2^2 \quad \dots(i)$$

Heat transfer through insulation is equal to heat transfer rate by convection to atmosphere. Therefore,

$$Q_2 = \frac{T_1 - T_2}{\frac{(r_2 - r_1)}{4\pi k r_2 r_1}}$$

$$= \frac{120 - 40}{\frac{r_2 - 0.5}{4\pi \times 20 \times 10^{-3} \times 0.5 r_2}} = \frac{10.05 r_2}{r_2 - 0.5} \dots (ii)$$

On equating equations (i) and (ii) we have,

$$7850 r_2^2 = \frac{10.05 r_2}{r_2 - 0.5}$$

$$7850 r_2^3 - 3925 r_2^2 - 10.05 r_2 = 0$$

$$7850 r_2^2 - 3925 r_2 - 10.05 = 0;$$

$$r_2^2 - 0.5 r_2 - 1.28 \times 10^{-3} = 0$$

By hit and trial ; $r_2 = 0.51 \text{ m}$

Thickness of insulation,

$$r_2 - r_1 = 0.51 - 0.5 = 0.01 \text{ m or 1cm}$$

...Ans.

(iii) Percentage reduction in heat loss

$$Q_2 = 7850 r_2^2 = 7850 \times 0.51^2 = 2041.8 \text{ W}$$

$$\% \text{ heat reduction} = \frac{Q_1 - Q_2}{Q_1} \times 100$$

$$= \frac{8242.5 - 2041.8}{8242.5} \times 100$$

$$= 75.2\%$$

...Ans.

Ex. 2.12.31 : A refrigerant suction line having outer diameter 30 mm is required to be thermally insulated. The outside air film coefficient of heat transfer is $12 \text{ W/m}^2 \text{ K}$. The thermal conductivity of insulation is 0.3 W/m K , (i) Determine whether the insulation will be effective; (ii) Estimate the maximum value of thermal conductivity of insulating material to reduce heat transfer; (iii) Determine the thickness of cork insulation to reduce the heat transfer to 22% if the thermal conductivity of cork is 0.038 W/mK .

Soln. :

$$d_1 = 30 \text{ mm} = 0.03 \text{ m},$$

$$\text{therefore, } r_1 = \frac{d_1}{2} = \frac{0.03}{2} = 0.015 \text{ m}$$

$$h = 12 \text{ W/m}^2 \text{ K},$$

Thermal conductivity of insulation, $k = 0.3 \text{ W/mK}$

(i) Whether insulation will be effective ?

Critical radius of insulation,

$$r_c = \frac{k}{h} = \frac{0.3}{12} = 0.025 \text{ m}$$

Since the critical radius is more than suction pipe radius, application of insulation will increase heat transfer rate rather than reducing the heat transfer. Therefore, **insulation is ineffective.** ...Ans.

(ii) Value of k_1 to reduce heat transfer rate

r_c should be equal to outer radius of the suction pipe.

$$\therefore r_c = r_1 = \frac{k_1}{h};$$

$$k_1 = r_1 \cdot h = 0.015 \times 12 = 0.18 \text{ W/mK} \dots \text{Ans.}$$

(iii) Thickness of cork insulation to reduce heat transfer to 22% if k_2 for cork is 0.038 W/mK (i)

$$\text{Given : } Q_2 = 0.22 Q_1$$

Heat transfer rate without insulation per meter length i.e. $L = 1 \text{ m}$

$$Q_1 = \frac{\Delta T}{R_1} = \frac{\Delta T}{\frac{1}{h \times 2\pi r_1 L}}$$

$$= \frac{\Delta T}{\frac{1}{12 \times 2 \times \pi \times 0.015 \times 1}}$$

$$= \frac{\Delta T}{\left(\frac{1}{131}\right)} \dots (ii)$$

Heat transfer rate per meter length with insulation of radius r_2

$$Q_2 = \frac{\Delta T}{\left[\frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi \times k_2 \times L} + \frac{1}{h \times 2\pi \times r_2 \times 1} \right]}$$

$$Q_2 = \frac{\Delta T}{\left[\frac{\ln\left(\frac{r_2}{0.015}\right)}{2\pi \times 0.038 \times 1} + \frac{1}{12 \times 2\pi \times r_2 \times 1} \right]}$$

$$Q_2 = \frac{\Delta T}{\left[\frac{\ln\left(\frac{r_2}{0.015}\right)}{2.388} + \frac{1}{75.4 r_2} \right]}$$

$$\text{But } Q_2 = 0.22 Q_1$$

$$\therefore \frac{\Delta T}{\left[\frac{\ln\left(\frac{r_2}{0.015}\right)}{2.388} + \frac{1}{75.4 r_2} \right]} = \frac{0.22 \times \Delta T}{\frac{1}{1.131}}$$

$$\frac{\ln(r_2 / 0.015)}{2.388} + \frac{1}{75.4 r_2} = \frac{1}{0.2488}$$

$$75.4 r_2 \ln\left(\frac{r_2}{0.015}\right) + 0.2388 = \frac{0.2388 \times 75.4 r_2}{0.2488}$$

$$75.4 r_2 \ln\left(\frac{r_2}{0.015}\right) - 72.37 r_2 + 0.2388 = 0$$

By trial and error method,

$$r_2 = 0.0421 \text{ m} = 42.1 \text{ mm}$$

∴ Thickness of insulation,

$$t = r_2 - r_1 = 42.1 - 15 = 27.1 \text{ mm} \quad \dots \text{Ans.}$$

Ex. 2.12.32 : An electrical conductor of 10 mm diameter, insulated by PVC ($k = 0.18 \text{ W/mK}$) is installed in air at 30°C having convective heat transfer coefficient of $7.8 \text{ W/m}^2\text{K}$. If the surface temperature of base conductor is 85°C , Calculate :

- Current carrying capacity of the conductor when 2 mm thick insulation is provided (resistivity of conductor $70 \mu\text{ohm-cm}$)
- Critical insulation thickness.
- Maximum current carrying capacity.
- Percentage increase in current carrying by providing critical insulation thickness.

Soln. :

Given : $d_1 = 10 \text{ mm}$,

$$r_1 = 5 \text{ mm} = 0.005 \text{ m},$$

$$\rho = 70 \mu\text{ Ohm-cm} = 70 \times 10^{-6} \times 10^{-2} \text{ Ohm} \cdot \text{m}$$

$$T_1 = 85^\circ\text{C}, \quad T_\infty = 30^\circ\text{C}, \quad h = 7.8 \text{ W/m}^2\text{K},$$

$$k = 0.18 \text{ W/mK}$$

$$r_2 = r_1 + \text{thickness of insulation}$$

$$= 0.005 + 0.002 = 0.007 \text{ m}.$$

(a) Current carrying capacity, I

$$Q = \frac{T_1 - T_\infty}{\frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k L} + \frac{1}{h \cdot 2\pi r_2 L}}$$

$$Q = \frac{85 - 30}{\frac{\ln\left(\frac{0.007}{0.005}\right)}{2\pi \times 0.18 \times 1} + \frac{1}{7.8 \times 2\pi \times 0.007 \times 1}}$$

$$= 17.12 \text{ W}$$

(Assuming length of conductor as 1 m)

$$Q = I^2 R$$

$$\therefore I = \sqrt{\frac{Q}{R}}$$

$$R = \rho \frac{L}{A} = \frac{70 \times 10^{-6} \times 10^{-2} \times 1}{\pi (0.005)^2} = 0.00892$$

$$I = \sqrt{\frac{17.12}{0.00892}} = 43.8 \text{ Amp.} \quad \dots \text{Ans.}$$

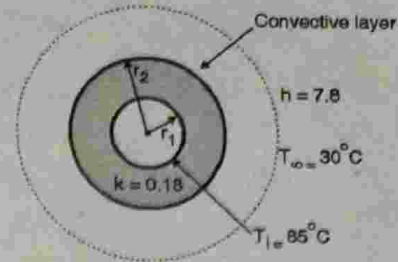


Fig. P. 2.12.32

(b) Critical radius of insulation

$$r_c = \frac{k}{h} = \frac{0.18}{7.8} = 0.023 \text{ m} = 23 \text{ mm}$$

Hence, critical thickness = $23 - 5 = 18 \text{ mm} \quad \dots \text{Ans.}$

(c) Maximum current carrying capacity

This will happen at $r_2 = r_c$

$$\therefore Q_{\max} = \frac{T_1 - T_\infty}{\frac{\ln\left(\frac{r_c}{r_1}\right)}{2\pi k L} + \frac{1}{h \cdot 2\pi r_c L}}$$

$$= \frac{85 - 30}{\frac{\ln\left(\frac{0.023}{0.005}\right)}{2\pi \times 0.18 \times 1} + \frac{1}{7.8 \times 2\pi \times 0.023 \times 1}}$$

$$= 24.56 \text{ W}$$

$$\therefore I_{\max} = \sqrt{\frac{Q_{\max}}{R}} = \sqrt{\frac{24.56}{0.00892}}$$

$$= 52.48 \text{ Amp.} \quad \dots \text{Ans.}$$

(d) Percentage increase in current carrying capacity

$$\frac{I_{\max} - I}{I} \times 100 = \frac{52.48 - 43.8}{43.8} \times 100$$

$$= 19.85\% \quad \dots \text{Ans.}$$

Ex. 2.12.33 : A steam pipe made of steel ($k = 58 \text{ W/m-K}$) has ID of 160 mm and OD of 170 mm. The saturated steam flowing through it is at 300°C , while the ambient air is at 50°C . It has two layers of insulation. The inner layer ($k = 0.17 \text{ W/m-K}$) is 30 mm thick and the outer layer ($k = 0.023 \text{ W/mK}$) is 50 mm thick. The heat transfer coefficient on inside and outside walls are $40 \text{ W/m}^2\text{K}$ and $5.8 \text{ W/m}^2\text{K}$ respectively. Find the rate of heat loss per unit length of the pipe.

Soln. :

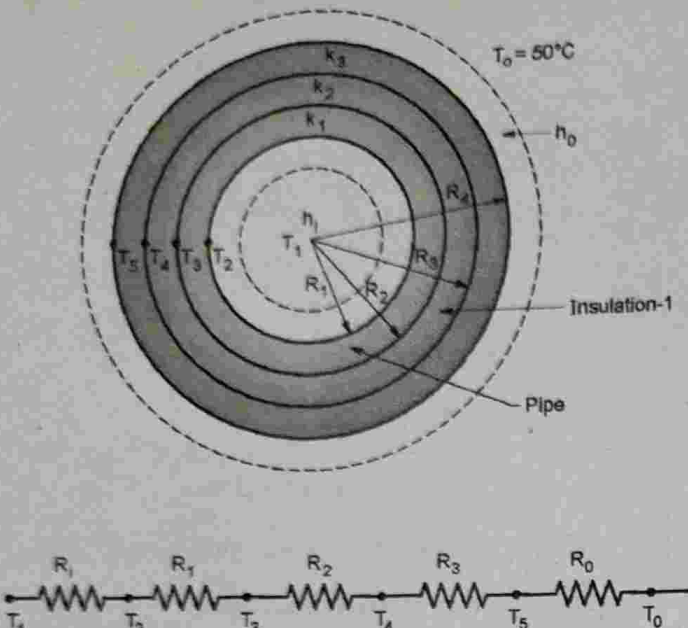


Fig. P. 2.12.33

$$d_1 = 160 \text{ mm} \therefore r_1 = \frac{d_1}{2} = \frac{160}{2} = 80 \text{ mm} = 0.08 \text{ m}$$

$$d_2 = 170 \text{ mm} \therefore r_2 = \frac{d_2}{2} = \frac{170}{2} = 85 \text{ mm} = 0.085 \text{ m}$$

$$r_3 = r_2 + 30 = 85 + 30 = 115 \text{ mm} = 0.115 \text{ m}$$

$$r_4 = r_3 + 50 = 115 + 50 = 165 \text{ mm} = 0.165 \text{ m}$$

Thermal conductivity of pipe, $k_1 = 58 \text{ W/mK}$

Thermal of inner insulation, $k_2 = 17 \text{ W/mK}$

Thermal of outer insulation $k_3 = 0.023 \text{ W/mK}$

Length of pipe, $L = 1 \text{ m}$

$T_1 = 300^\circ\text{C}$, $h_i = 40 \text{ W/m}^2\text{K}$

$T_0 = 50^\circ\text{C}$, $h_o = 5.8 \text{ W/m}^2\text{K}$

(i) Heat transfer rate (Q)

Resistance of steel pipe, R_1

$$\therefore R_1 = \frac{\log_e(r_2/r_1)}{2\pi L k_1} = \frac{\log_e(85/80)}{2\pi \times 1 \times 58} = 1.66357 \times 10^{-4} \text{ K/W}$$

Resistance of inner insulation, R_2

$$R_2 = \frac{\log_e(r_3/r_2)}{2\pi L k_2} = \frac{\log_e(155/85)}{2\pi \times 1 \times 0.17} = 0.28299 \text{ K/W}$$

Resistance of outer insulation, R_3

$$R_3 = \frac{\log_e(r_4/r_3)}{2\pi L k_3} = \frac{\log_e(165/115)}{2\pi \times 1 \times 0.023}$$

$$= 2.4981 \text{ K/W}$$

Convective resistance at inside, R_i

$$R_i = \frac{1}{h_i \pi d_1 L} = \frac{1}{h_i 2\pi r_1 L} = \frac{1}{40 \times 2\pi \times 0.08 \times 1} = 0.04973 \text{ K/W}$$

Convective resistance at outside, R_o

$$R_o = \frac{1}{h_o \pi d_4 L} = \frac{1}{h_o \times 2\pi \times r_4 \times L} = \frac{1}{5.8 \times 2\pi \times 0.165 \times 1} = 0.1663 \text{ K/W}$$

$$\therefore \Sigma R = R_i + R_2 + R_3 + R_1 + R_o = 1.66357 \times 10^{-4} + 0.28299 + 2.4981 + 0.04973 + 0.1663 = 2.997 \text{ K/W}$$

$$\therefore Q = \frac{\Delta T}{\Sigma R} = \frac{300 - 50}{2.997}$$

$$= 83.4 \text{ W/m length}$$

...Ans.

Ex. 2.12.34 : A steel tube of 5 cm inner diameter and 8 cm outer diameter ($k = 16 \text{ W/mK}$), is covered with an insulation of 3 cm thickness ($k = 0.3 \text{ W/mK}$). A hot gas at 350°C $h = 400 \text{ W/m}^2\text{K}$ flows. Calculate the heat loss from the tube for 20 meter length. Also calculate the temperature at the interface of insulation and steel.

Soln. : Refer Fig. P. 2.12.34.

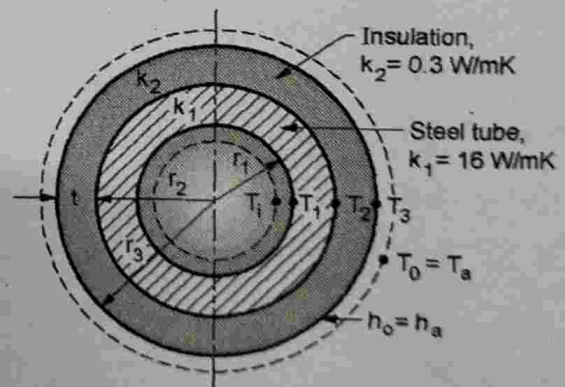


Fig. P. 2.12.34

Steel tube : Inner diameter, $d_1 = 5 \text{ cm}$

i.e. $r_1 = 2.5 \text{ cm} = 0.025 \text{ m}$

$d_2 = 8 \text{ cm}$, $r_2 = \frac{8}{2} = 4 \text{ cm} = 0.04 \text{ m}$, $k_1 = 16 \text{ W/mK}$

Insulation : Thickness, $t = 3 \text{ cm}$

$r_3 = r_2 + t = 4 + 3 = 7 \text{ cm} = 0.07 \text{ m}$

$k_2 = 0.3 \text{ W/mK}$

Gas. $T_g = T_i = 350^\circ\text{C}$, $h_i = 400 \text{ W/m}^2\text{K}$

$$L = 20 \text{ m}$$

Air. $T_a = T_o = 30^\circ\text{C}$, $h_o = h_a = 60 \text{ W/m}^2\text{K}$

Thermal resistance s are as follows :

Gas, $R_i = \frac{1}{h_i \cdot A_i \cdot L} = \frac{1}{h_i \times 2\pi r_1 L}$

$$= \frac{1}{400 \times 2 \times \pi \times 0.025 \times 20}$$

$$= 7.958 \times 10^{-4} \text{ K/W}$$

Steel tube, $R_1 = \frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi L k_1} = \frac{\ln\left(\frac{4}{2.5}\right)}{2\pi \times 20 \times 16}$

$$= 2.338 \times 10^{-4} \text{ K/W}$$

Insulation, $R_2 = \frac{\ln(r_3/r_2)}{2\pi L k_2} = \frac{\ln(7/4)}{2\pi \times 20 \times 0.3}$

$$= 148.443 \times 10^{-4} \text{ K/W}$$

Air, $R_o = \frac{1}{h_o \cdot A_o \cdot L} = \frac{1}{h_o \times 2\pi r_3 \times L}$

$$= \frac{1}{60 \times 2\pi \times 0.07 \times 20}$$

$$= 18.947 \times 10^{-4} \text{ K/W}$$

$$\Sigma R = R_i + R_1 + R_2 + R_o$$

$$= (7.958 + 2.338 + 148.443 + 18.947) \times 10^{-4}$$

$$= 177.686 \times 10^{-4} \text{ K/W}$$

1. Heat loss form tube, Q

$$Q = \frac{(T_i - T_o)}{\Sigma R} = \frac{(350 - 30)}{177.686 \times 10^{-4}}$$

$$= 18.01 \times 10^3 \text{ W} = 18.01 \text{ kW} \quad \dots\text{Ans.}$$

2. Temperature at the interface of insulation and tube, T_2

$$Q = \frac{(T_i - T_2)}{R_i + R_1}$$

$$18.01 \times 10^3 = \frac{(350 - T_2)}{(7.958 \times 10^{-4} + 2.338 \times 10^{-4})}$$

$$T_2 = 331.46^\circ\text{C} \quad \dots\text{Ans.}$$

Ex. 2.12.35 : A standard cast iron pipe (inner diameter = 50 mm and outer diameter = 55 mm) is insulated with magnesium insulation ($k = 0.02 \text{ W/mK}$). Temperature at the interface between the pipe and insulation is 300°C . The allowable heat loss through the pipe is 600 W/m length of pipe and for the safety, the temperature of the outside surface of insulation must not exceed 100°C . Determine :

- Minimum thickness of insulation required and
- The temperature of inside surface of the pipe assuming its thermal conductivity 20 W/mK .

Soln. :

(i) Minimum thickness of insulation

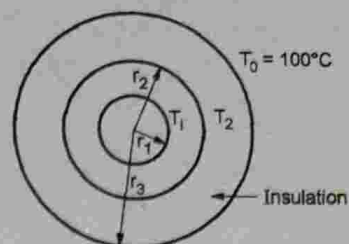


Fig. P. 2.12.35

$$T_o = 100^\circ, \quad T_2 = 300^\circ\text{C}$$

$$r_2 = \frac{55}{2} = 0.0275 \text{ m}$$

$$\text{Now } Q = \frac{T_2 - T_o}{R}$$

Where $R = \frac{\ln\left[\frac{r_2}{r_1}\right]}{2\pi k L}$

$$\therefore 600 = \frac{(300 - 100) \times 2 \times \pi \times 0.02}{\ln\left[\frac{r_3}{0.0275}\right]}$$

$$\therefore \ln\left[\frac{r_3}{0.0275}\right] = 0.041$$

$$\frac{r_3}{0.0275} = 1.042$$

$$r_3 = 28.65 \text{ mm}$$

$$\therefore \text{Thickness required} = r_3 - r_2 = 28.65 - 27.5$$

$$= 1.15 \text{ mm} \quad \dots\text{Ans.}$$

(ii) When k for pipe = 20 W/mK , $T_i = ?$

$$Q = \frac{T_i - T_o}{\frac{\ln\left[\frac{r_2}{r_1}\right]}{2\pi k_1} + \frac{\ln\left[\frac{r_3}{r_2}\right]}{2\pi k_2}}$$

$$600 = \frac{T_i - 100}{\frac{\ln\left[\frac{0.0275}{0.0250}\right]}{2\pi \times 20} + \frac{\ln\left[\frac{0.0286}{0.0275}\right]}{2\pi \times 0.02}}$$

$$\therefore T_i = 287^\circ\text{C} \quad \dots\text{Ans.}$$

Ex. 2.12.36 : Steam pipe made of steel ($k = 58 \text{ W/m-K}$) has ID of 160 mm and, OD of 170 mm. The saturated steam flowing through it is at 300°C , while the ambient air is at 50°C . It has two layers of insulation. The inner layer ($k = 0.17 \text{ W/m-K}$) is 30 mm thick and the outer layer ($k = 0.023 \text{ W/mK}$) is 50 mm thick. The heat transfer coefficient on inside and outside walls are $40 \text{ W/m}^2\text{K}$ and $5.8 \text{ W/m}^2\text{K}$ respectively. Find the rate of heat loss per unit length of the pipe.



Soln. :

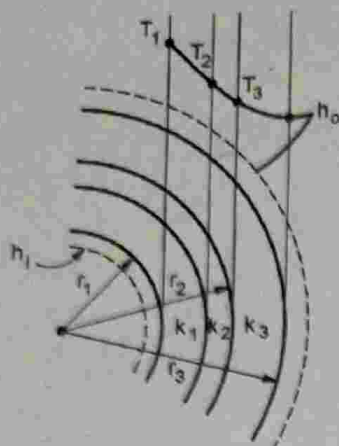


Fig. P. 2.12.36

There are a number of thermal resistances in series through which heat flows from the steam pipe to the ambient air. The rate of heat flow per unit length

$$\begin{aligned} \frac{Q}{L} &= \frac{T_1 - T_0}{R_1 + R_2 + R_3 + R_4 + R_5} \\ &= \frac{2\pi (T_1 - T_0)}{\frac{1}{h_1 r_1} + \frac{\ln\left(\frac{r_2}{r_1}\right)}{K_{w1}} + \frac{\ln\left(\frac{r_3}{r_2}\right)}{K_{w2}} + \frac{\ln\left(\frac{r_4}{r_3}\right)}{K_{w3}} + \frac{1}{h_0 r_4}} \end{aligned}$$

Where,

$$\begin{aligned} r_1 &= 80 \text{ mm}, & r_2 &= 85 \text{ mm} \\ r_3 &= 115 \text{ mm} & \text{and } r_4 &= 165 \text{ mm} \end{aligned}$$

$$\begin{aligned} \frac{Q}{L} &= \frac{2\pi (300 - 50)}{\frac{1}{40 \times 0.08} + \frac{\ln\left(\frac{85}{80}\right)}{58} + \frac{\ln\left(\frac{115}{85}\right)}{0.17} + \frac{\ln\left(\frac{165}{115}\right)}{0.023} + \frac{1}{58 \times 0.16}} \\ &= \frac{500\pi}{0.3125 + 0.00105 + 1.738 + 15.69 + 0.108} \\ &= 88 \text{ W/m} \end{aligned}$$

...Ans.

Ex. 2.12.37 : A 1 mm diameter wire is maintained at a temperature of 400°C and exposed to a convection environment at 40°C with $h = 20 \text{ W/m}^2\text{C}$. Calculate the thermal conductivity which will cause an insulation of 0.2 mm to produce a "critical radius". How much of this insulation must be added to reduce the heat transfer by 75% from that which would be experienced by the bare wire?

Soln. :

Wire diameter, $d_1 = 1 \text{ mm} = 1 \times 10^{-3} \text{ m}$; $T_s = 400^\circ\text{C}$,

$$T_\infty = 40^\circ\text{C}; h = 120 \text{ W/m}^2\text{C};$$

Critical radius, $r_c = 0.5 \text{ mm} + 0.2 \text{ mm} = 0.7 \times 10^{-3} \text{ m}$

1. Thermal conductivity of wire, k

Considering wire of length 1 m.

$$\text{Critical radius, } r_c = \frac{k}{h}; \quad 0.7 \times 10^{-3} = \frac{k}{120}$$

$$\therefore k = 0.084 \text{ W/mK}$$

...Ans.

2. Thickness of insulation to be added to reduce heat transfer rate by 75% as against the heat transfer rate by bare wire

Heat transfer rate through bare wire

$$\begin{aligned} Q &= h \cdot A \times (T_s - T_\infty) = h \cdot \pi d_1 L (T_s - T_\infty) \\ &= 120 \times \pi \times (1 \times 10^{-3}) \times 1 \times (400 - 40) \\ &= 135.72 \text{ W} \end{aligned}$$

$$Q' = (1 - 0.75) Q = 0.25 \times 135.72 = 33.93 \text{ W}$$

$$Q' = \frac{(T_s - T_\infty)}{\frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k L} + \frac{1}{h \cdot 2\pi r_2 \cdot L}}$$

$$33.93 = \frac{(400 - 40)}{\left[\frac{\ln(r_2/0.0005)}{2\pi \times 0.084 \times 1} + \frac{1}{120 \times 2\pi \times r_2 \times 1} \right]}$$

By hit and trial; $r_2 = 0.135 \text{ m}$

...Ans.

Ex. 2.12.38 : A hot gas at 330°C with convection coefficient $222 \text{ W/m}^2 \text{K}$ is flowing through a steel tube of outside diameter 8 cm and thickness 1.3 cm. It is covered with an insulating material of thickness 2 cm, having conductivity of 0.2 W/mK . The outer surface of insulation is exposed to ambient air at 25°C with convection coefficient of $55 \text{ W/m}^2 \text{K}$.

Calculate :

(i) Heat loss to air from 5 m long tube.

(ii) The temperature drop due to thermal resistance of hot gases, steel tube, the insulation layer and the outside air.

Take conductivity of steel = 50 W/mK .

Soln. : Refer Fig. P. 2.17.47.

$$\begin{aligned} \text{Gas :} \quad T_1 &= T_{\text{gas}} = 330^\circ\text{C}, \\ h_1 &= h_g = 222 \text{ W/m}^2 \text{K} \end{aligned}$$

$$\text{Steel tube :} \quad r_2 = \frac{d_2}{2} = \frac{8}{2} = 4 \text{ cm, } t = 1.3 \text{ cm};$$

$$r_1 = r_2 - t = 4 - 1.3 = 2.7 \text{ cm}$$

$$\begin{aligned} \text{Insulation :} \quad t_1 &= 2 \text{ cm; } r_3 = r_2 + t_1 = 4 + 2 = 6 \text{ cm,} \\ k_2 &= 0.2 \text{ W/mK} \end{aligned}$$

$$\text{Air :} \quad T_0 = 25^\circ\text{C}, \quad h_0 = h_{\text{air}} = 55 \text{ W/m}^2 \text{K}$$

Length of tube, $L = 5 \text{ m}$

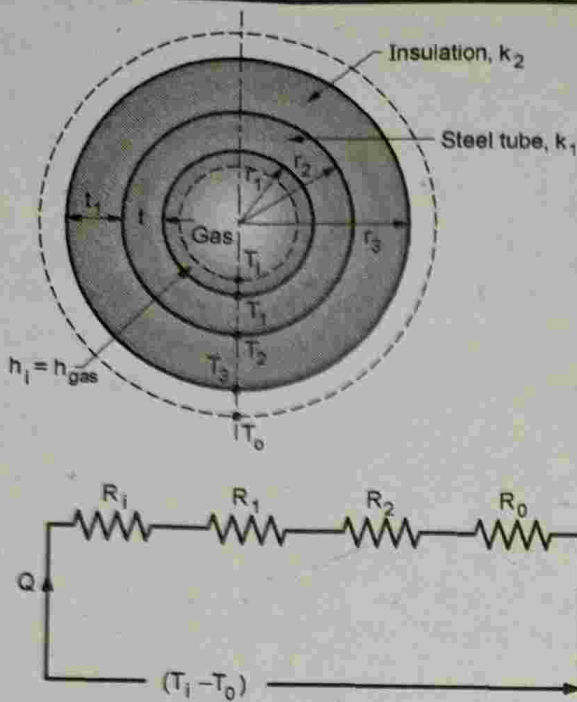


Fig. P. 2.12.38

Resistances are :

$$R_i = \frac{1}{h_i \cdot A_i} = \frac{1}{h_i \times 2\pi r_1 L} = \frac{1}{222 \times 2 \times \pi (2.7/100) 5} = 5.31 \times 10^{-3} \text{ K/W}$$

$$R_1 = \frac{\ln(r_2/r_1)}{2\pi k_1 L} = \frac{\ln(4/2.7)}{2\pi \times 50 \times 5} = 0.25 \times 10^{-3} \text{ K/W}$$

$$R_2 = \frac{\ln(r_3/r_2)}{2\pi L k_2} = \frac{\ln(6/4)}{2\pi \times 50 \times 0.2} = 6.453 \times 10^{-3} \text{ K/W}$$

$$R_0 = \frac{1}{h_0 A_0} = \frac{1}{h_0 \times 2\pi r_3 L} = \frac{1}{55 \times 2 \times \pi \times (6 \times 10^{-2}) \times 5} = 9.646 \times 10^{-3} \text{ K/W}$$

$$\begin{aligned} \therefore \Sigma R &= R_i + R_1 + R_2 + R_0 \\ &= (5.31 \times 10^{-3} + 0.25 \times 10^{-3} + 6.453 \times 10^{-3} + 9.646 \times 10^{-3}) \\ &= 21.659 \times 10^{-3} \text{ K/W} \end{aligned}$$

(i) Heat loss to air, Q

$$Q = \frac{T_1 - T_0}{\Sigma R} = \frac{(330 - 25)}{21.659 \times 10^{-3}} = 14081.9 \text{ W} \quad \dots \text{Ans.}$$

(ii) Temperature drop due to thermal resistance for the following cases

$$Q = \frac{T_1 - T_1}{R_i} = \frac{T_1 - T_2}{R_1} = \frac{T_2 - T_3}{R_2} = \frac{T_3 - T_0}{R_0} \quad \dots (i)$$

(a) Gases ($T_1 - T_1$)

From Equation (i),

$$\begin{aligned} (T_1 - T_1) &= Q \times R_i = 14081.9 \times (5.31 \times 10^{-3}) \\ &= 74.77^\circ\text{C} \end{aligned}$$

...Ans.

(b) Steel tube, ($T_1 - T_2$)

$$\begin{aligned} (T_1 - T_2) &= Q \times R_1 \\ &= 14081.9 \times (0.25 \times 10^{-3}) = 3.52^\circ\text{C} \end{aligned}$$

...Ans.

(c) The insulation, ($T_2 - T_3$)

$$\begin{aligned} (T_2 - T_3) &= Q \times R_2 \\ &= 14081.9 \times (6.453 \times 10^{-3}) \\ &= 90.87^\circ\text{C} \end{aligned}$$

...Ans.

(d) Outside air, ($T_3 - T_0$)

$$\begin{aligned} (T_3 - T_0) &= Q \times R_0 = 14081.9 \times (9.646 \times 10^{-3}) \\ &= 135.84^\circ\text{C} \end{aligned}$$

...Ans.

Ex. 2.12.39 : A 60 W lamp is buried in soil ($k = 0.83 \text{ W/mK}$) at 0°C and switched on. Find the temperature 0.3m away from the lamp, when steady state is reached.

Soln. : $Q = 60 \text{ W}$. The heat is dissipated by lamp considering it as spherical in shape through the soil. Let T_1 be the temperature at 0.3 m radius from Fourier's law,

$$Q = -k \cdot A \cdot \frac{dT}{dr} = -k \cdot (4\pi r^2) \cdot \frac{dT}{dr}$$

$$Q \cdot \frac{dr}{r^2} = -k \cdot 4\pi \cdot dT \quad \dots (i)$$

Boundary conditions are :

$$\text{At } r = 0.3;$$

$$T = T_1$$

$$\text{At } r = \infty;$$

$$T = 0$$

On integrating equation (i) within boundary conditions,

$$\int_{0.3}^{\infty} \left(\frac{dr}{r^2} \right) = -k \cdot 4\pi \int_{T_1}^0 dT$$

$$Q \left[-\frac{1}{r} \right]_{0.3}^{\infty} = -4\pi k \cdot [T]_{T_1}^0$$

$$60 \left[-\frac{1}{\infty} + \frac{1}{0.3} \right] = -4\pi \times 0.83 (0 - T_1)$$

$$T_1 = 19.175^\circ\text{C} \quad \dots \text{Ans.}$$

Ex. 2.12.40 : A pipe of outside diameter 20 mm is to be insulated with asbestos which has a mean thermal conductivity of 0.1 W/mK . The local coefficient of connectivity heat transfer is $5 \text{ W/m}^2\text{K}$. Comment upon the suitability of asbestos as the insulating material. What should be the minimum value of thermal conductivity of insulating material to reduce heat transfer?

Soln. : $d_o = 20 \text{ mm}$ i.e. $r_o = \frac{20}{2} = 10 \text{ mm}$

$$k = 0.1 \text{ W/mK}, \quad h = 5 \text{ W/m}^2 \text{ K}$$

The critical radius of insulation for optimum heat transfer is given by,

$$r_c = \frac{k}{h_o} = \frac{0.1}{5} = 0.02 \text{ m} \quad \text{or} \quad 20 \text{ mm}$$

1. Effectiveness of applied insulating material

In order to reduce the heat transfer rate then outside radius of tube r_o should be greater than r_c .

But, in this problem, $r_o (= 10 \text{ mm}) < r_c (= 20 \text{ mm})$. Hence, the applied insulating material is not suitable.

...Ans.

2. Minimum value of thermal conductivity of insulating material to reduce heat transfer, k_1

The condition is,

$$r_o \geq r_c = \frac{k_1}{h_o} \quad \text{i.e.} \quad 0.01 = \frac{k_1}{5}$$

$$k_1 = 0.05 \text{ W/m K}$$

...Ans.

Thus the maximum conductivity of insulating material permitted is **0.05 W/m K**.

...Ans.

Ex. 2.12.41 : A cylindrical electrical conductor of 10 mm diameter, insulated by plastic ($k = 0.16 \text{ W/mK}$) is located in the air at 25°C having convective heat transfer coefficient of $8 \text{ W/m}^2 \text{ K}$. If the surface temperature of the base conductor is 90°C . Calculate; under steady state condition,

(i) Current passing through conductor when 2 mm thick insulation is provided. Electrical resistance per meter length of the conductor is $9 \times 10^{-3} \text{ ohm}$.

(ii) Current carrying capacity of the conductor when the insulation of critical radius is provided over it.

Soln. : Refer Fig. P. 2.12.41.

Given : Radius of electrical wire, $r_1 = \frac{d_1}{2} = \frac{10}{2} = 5 \text{ mm}$

$$= 0.005 \text{ m}$$

Outer radius with insulation,

$$r_2 = r_1 + \text{thickness} = (5 + 2) \text{ mm} = 0.007 \text{ m}$$

Surface temperature of conductor, $T = 90^\circ\text{C}$

Air,
 $h_a = 8 \text{ W/m}^2 \text{ K}$
 $T_a = 25^\circ\text{C}$

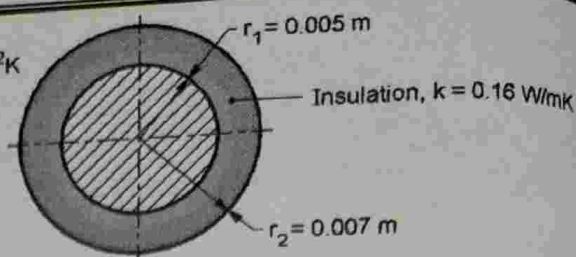


Fig. P. 2.12.41

Assume, $L = 1 \text{ m}$

Radial heat flow through a composite cylinder

$$Q = \frac{T - T_a}{\frac{1}{2\pi r_2 \cdot h_a \cdot L} + \frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k L}} \quad \dots(i)$$

$$= \frac{(90 - 25)}{\frac{1}{2\pi \times 0.007 \times 8 \times 1} + \frac{\ln\left(\frac{0.007}{0.005}\right)}{2\pi \times 0.16 \times 1}}$$

$$= 20.46 \text{ W}$$

(i) Current passing through the conductor, I if resistance, $R = 9 \times 10^{-3} \text{ ohm}$

$$Q = I^2 R \quad I = \sqrt{\frac{Q}{R}}$$

$$I = \sqrt{\frac{20.46}{9 \times 10^{-3}}} = 47.68 \text{ amp} \quad \dots\text{Ans.}$$

(ii) Current carrying capacity of conductor when critical insulation is provided, I_1

$$\text{Critical radius, } r_c = \frac{k}{h} = \frac{0.16}{8} = 0.02 \text{ m.} = r_2$$

From Equation (i) above,

$$Q_1 = \frac{(90 - 25)}{\left[\frac{1}{2\pi \times 0.02 \times 8 \times 1} + \frac{\ln(0.02/0.005)}{2\pi \times 0.16 \times 1} \right]}$$

$$= 27.38 \text{ W}$$

$$I_1 = \sqrt{\frac{Q_1}{R}} = \sqrt{\frac{27.38}{9 \times 10^{-3}}} = 55.16 \text{ amp} \quad \dots\text{Ans.}$$

Ex. 2.12.42 : A 2.5 cm outer diameter refrigerant suction line is required to be insulated. Outside heat transfer coefficient is $9.5 \text{ W/m}^2 \text{ K}$. Thermal conductivity of insulation is 0.18 W/mK .

(i) Verify whether insulation would be effective or not.

(ii) What will be maximum value of thermal conductivity of insulation to reduce heat transfer?

(iii) What should be the thickness of cork insulation to reduce heat transfer to 20% if the thermal conductivity of cork is 0.04 W/mK ?

Soln. :

Given : $D = 2.5 \text{ cm} = 0.025 \text{ m}$, $r_1 = 0.0125 \text{ m}$
 $h = 9.5 \text{ W/m}^2\text{K}$, $k = 0.18 \text{ W/mK}$.

(i) Whether insulation would be effective ?

$$r_c = \frac{k}{h} = \frac{0.18}{9.5} = 0.01895 \text{ m}$$

Since critical radius is more than suction pipe radius, application of insulation will increase heat transfer rate rather than reducing it. So insulation is ineffective.

(ii) $k = ?$ to reduce the heat transfer rate.

r_c should be equal to outer radius of the suction pipe.

$$r_c = r_1 = \frac{k}{h}$$

$$k = r_1 h = 0.0125 \times 9.5$$

$$= 0.11875 \text{ W/mK (max value of } k) \text{ ...Ans.}$$

(iii) $Q_{II} = 0.2 Q_I$ for cork $k = 0.04 \text{ W/mK}$

Heat transfer without insulation

$$Q_I = \frac{\frac{\Delta T}{1}}{9.5 \times 2\pi \times 0.0125} = \frac{\frac{\Delta T}{1}}{0.74575}$$

With circle insulation,

$$Q_{II} = \frac{\Delta T}{\frac{\ln\left(\frac{r_2}{0.0125}\right)}{2\pi \times 0.04 \times 1} + \frac{1}{9.5 \times 2\pi \times r_2 \times 1}}$$

$$= \frac{\Delta T}{\frac{\ln\left(\frac{r_2}{0.0125}\right)}{0.2512} + \frac{1}{59.66 r_2}}$$

For heat transfer reduction to 20%,

$$Q_{II} = 0.2 Q_I$$

$$\frac{\frac{\Delta T}{\ln\left(\frac{r_2}{0.0125}\right) + \frac{1}{59.66 r_2}}}{0.2512} = \frac{0.2 \times \Delta T}{0.74575}$$

$$\therefore \frac{\ln\left(\frac{r_2}{0.0125}\right)}{0.2512} + \frac{1}{59.66 r_2} = \frac{1}{0.1492}$$

$$59.66 r_2 \ln\left(\frac{r_2}{0.0125}\right) + 0.2512 = 100.45 r_2$$

By trial and error, $r_2 = 0.064 \text{ mm}$...Ans.

Ex. 2.12.43: The hot gas at 325°C flows through a metal pipe 10 cm. O.D. and 3 mm. thick. It is insulated with mineral wool in such a way that outside of insulation temperature should not exceed 50°C . Determine the thickness of insulation taking following data.

$$h_i = 25 \text{ W/m}^2\text{K}, \quad h_o = 10 \text{ W/m}^2\text{K},$$

$$T_a = (\text{surrounding air temperature}) = 25^\circ\text{C}$$

$$K (\text{metal pipe}) = 40 \text{ W/mK}, \quad K (\text{mineral wool}) = 0.045 \text{ W/mK}.$$

Also calculate the loss of heat per meter after putting the insulation.

Soln. :

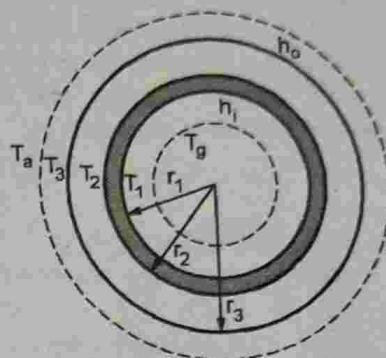


Fig. P. 2.12.43

$$r_2 = 5 \text{ cm} = 0.05 \text{ m}$$

$$r_1 = 50 \text{ mm} - 3 \text{ mm} = 47 \text{ mm} = 0.047 \text{ m}$$

$$r_3 = ?$$

$$T_g = 325^\circ\text{C}; \quad T_3 = 50^\circ\text{C}$$

$$T_a = 25^\circ\text{C}$$

$$k_m = 40 \text{ W/mK}; \quad k_w = 0.045 \text{ W/mK};$$

$$h_i = 25 \text{ W/m}^2\text{K}, \quad h_o = 10 \text{ W/m}^2\text{K}$$

Heat flow rate per meter length of cylinder can be written as

$$Q = \frac{T_g - T_3}{\frac{1}{h_i A_i} + \frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k_m \times L} + \frac{\ln\left(\frac{r_3}{r_2}\right)}{2\pi k_w L}} = \frac{T_3 - T_a}{\frac{1}{h_o A_o}}$$

$$= \frac{325 - 50}{\frac{1}{25 \times 2\pi \times 0.047} + \frac{\ln\left(\frac{0.05}{0.047}\right)}{2\pi \times 40 \times 1} + \frac{\ln\left(\frac{r_3}{0.05}\right)}{2\pi \times 0.045 \times 1}}$$

$$= \frac{50 - 25}{\frac{1}{10 \times 2\pi \times r_3}}$$

$$\frac{4.379}{r_3} = 25 \left[0.1355 + 0.000246 + \frac{\ln\left(\frac{r_3}{0.05}\right)}{0.2826} \right]$$

$$= 3.387 + 0.00615 + 88.46 \ln \left(\frac{r_3}{0.05} \right)$$

$$\frac{4.379}{r_3} - 88.46 \ln \left(\frac{r_3}{0.05} \right) - 3.387 = 0$$

By trial and error, $r_3 = 0.086 \text{ m} = 86 \text{ mm}$

Thickness of insulation, $= 86 - 50 = 36 \text{ mm} \quad \dots \text{Ans.}$

Ex. 2.12.44 : An insulated heat pipe of 16 cm diameter is covered with 4 cm thick layer of insulation ($k = 0.9 \text{ W/m}^\circ\text{C}$) and carries process steam. Determine the percentage change in the rate of heat loss if an extra 2 cm thick layer of lagging ($k = 1.25 \text{ W/m}^\circ\text{C}$) is provided. Given that surrounding temperature remains constant and the outer surface heat transfer coefficient for both the configurations is $12 \text{ W/m}^2\text{C}$.

Soln. :

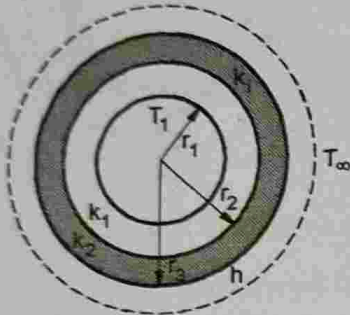


Fig. P. 2.12.44

Given : $D = 16 \text{ cm}$, $r_1 = 8 \text{ cm} = 0.08 \text{ m}$
 $r_2 = 8 \text{ cm} + 4 \text{ cm} = 12 \text{ cm} = 0.12 \text{ m}$
 $r_3 = 12 + 2 = 14 \text{ cm} = 0.14 \text{ m}$
 $k_1 = 0.9 \text{ W/mK}$; $k_2 = 1.25 \text{ W/m}^\circ\text{C}$
 $h = 12 \text{ W/m}^2\text{C}$

(i) Heat transfer rate without 2nd lagging (k_2)

$$Q_1 = \frac{T_1 - T_\infty}{\ln \frac{r_2}{r_1} + \frac{1}{h \times 2\pi r_2 L}}$$

$$= \frac{\Delta T}{\frac{\ln \left(\frac{0.12}{0.08} \right)}{2\pi \times 0.9 \times 1} + \frac{1}{12 \times 2\pi \times 0.12 \times 1}}$$

$$= \frac{\Delta T}{0.072 + 0.111} = 5.477 \Delta T$$

(ii) Heat transfer rate with 2nd lagging :

$$Q_2 = \frac{\Delta T}{\frac{\ln \left(\frac{r_2}{r_1} \right)}{2\pi \times k_1 L} + \frac{\ln \left(\frac{r_3}{r_2} \right)}{2\pi \times k_2 L} + \frac{1}{h \times 2\pi r_3 L}}$$

$$= \frac{\Delta T}{0.072 + \frac{\ln \left(\frac{0.14}{0.12} \right)}{2\pi \times 1.25 \times 1} + \frac{1}{12 \times 2\pi \times 0.14 \times 1}}$$

$$= \frac{\Delta T}{0.072 + 0.0196 + 0.095} = 5.365 \Delta T$$

$$\text{Percentage reduction} = \frac{Q_1 - Q_2}{Q_1} \times 100$$

$$= \frac{5.477 \Delta T - 5.365 \Delta T}{5.477 \Delta T} \times 100$$

$$= 2\% \quad \dots \text{Ans.}$$

Ex. 2.12.45 : The inner and outer radii of a hollow cylinder are 50 mm and 100 mm respectively. The inside surface is maintained at 300°C and the outer surface at 100°C . The thermal conductivity varies with the temperature over this temperature range as $k(T) = 0.5 + 0.5 \times 10^{-3} T$, where T is in $^\circ\text{C}$ and $k(T)$ is in $\text{W/m}^\circ\text{C}$. Determine.

- Heat flow rate per meter length of cylinder.
- Temperature at mid thickness of the shell.

Soln. :

Given : $r_1 = 50 \text{ mm} = 0.05 \text{ m}$
 $r_2 = 100 \text{ mm} = 0.1 \text{ m}$
 $T_1 = 300^\circ\text{C}$
 $T_2 = 100^\circ\text{C}$
 $k(T) = 0.5 + 0.5 \times 10^{-3} T$

(i) By Fourier's law

$$Q = k \cdot A \frac{dT}{dr} = -k \cdot 2\pi r L \cdot \frac{dT}{dr}$$

$$\text{or } \frac{Q}{2\pi L} \cdot \frac{dr}{r} = -k dT = -[0.5 + 0.5 \times 10^{-3} \cdot T] dT$$

Boundary conditions

At $r = r_1$; $T = T_1$
 At $r = r_2$; $T = T_2$

Integrating equation

$$\frac{Q}{2\pi L} \int_{r_1}^{r_2} \frac{dr}{r} = - \int_{T_1}^{T_2} (0.5 + 0.5 \times 10^{-3} T) dT$$

$$\frac{Q}{2\pi L} [\ln r]_{r_1}^{r_2} = - \left[0.5 T + 0.5 \times 10^{-3} \cdot \frac{T^2}{2} \right]_{T_1}^{T_2}$$

$$\frac{Q}{L} = - \frac{2\pi}{\ln \left(\frac{r_2}{r_1} \right)} \left[0.5 (T_2 - T_1) + 0.25 \times 10^{-3} (T_2^2 - T_1^2) \right]$$

$$= -\frac{2\pi}{\ln\left(\frac{0.1}{0.05}\right)} [0.5(100-300) + 0.25 \times 10^{-3}(100^2 - 300^2)]$$

$$= 9[100 + 20]$$

$$= 1080 \text{ W/m from inside to outside}$$

...Ans.

(ii) For mid temperature, T_m

Boundary condition

$$\text{At } r = r_1; \quad T = T_1$$

$$\text{At } r = r_m; \quad T = T_m$$

$$r_m = 0.075 \text{ m}; \quad \frac{Q}{L} = 1080$$

$$= -\frac{2\pi}{\ln\left(\frac{0.075}{0.05}\right)}$$

$$[0.5(T_m - 300) + 0.25 \times 10^{-3}(T_m^2 - 300^2)]$$

$$69.73 = -[0.5 T_m - 150 + 0.25 \times 10^{-3} T_m^2 - 22.5]$$

$$= -[0.25 \times 10^{-3} T_m^2 + 0.5 T_m - 172.5]$$

$$0.25 \times 10^{-3} T_m^2 + 0.5 T_m - 102.77 = 0$$

$$T_m^2 + 2000 T_m - 411080 = 0$$

$$T_m = \frac{-2000 \pm \sqrt{2000^2 - 4 \times 1 \times (-411080)}}{2}$$

$$= \frac{-2000 \pm 2375.78}{2} = 187.9^\circ \text{C} \quad \dots \text{Ans.}$$

Summary

Section I : Overall Heat Transfer Coefficient

Overall heat transfer coefficient, U

Heat flow rate through a composite wall by all modes of heat transfer (i.e. conduction, convection and radiation) can be expressed as :

$$Q = U \cdot A (T_1 - T_o) = \frac{T_1 - T_o}{\left(\frac{1}{UA}\right)} = \frac{T_1 - T_o}{\sum R}$$

$$\frac{1}{U} = \frac{1}{h_i} + \frac{x_1}{k_1} + \frac{x_2}{k_2} + \dots + \frac{1}{h_o}$$

Where, U is called overall heat transfer coefficient and $\frac{1}{UA}$ is called combined thermal resistance to composite wall.

(i) Resistance in series : $\sum R = R_1 + R_2 + R_3 + \dots$

(ii) Resistance in parallel : $\frac{1}{\sum R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$

Conductive Resistance, $R = \frac{x}{k \cdot A}$, convective resistance

$$R = \frac{1}{h \cdot A}$$

Estimation of intermediate temperature in a composite wall

$$Q = \frac{T_1 - T_o}{\sum R} = hA (T_1 - T_i) = \frac{k_1 A}{x_1} (T_1 - T_2)$$

$$= \frac{k_2 \cdot A}{x_2} (T_2 - T_3) = h_o \cdot A (T_3 - T_o)$$

At interface of composite wall, there is sudden drop in temperature due to thermal contact.

Resistance due to filling of air/fluid when at the interface the surfaces do not fit lightly.

Heat conduction through hollow long cylinder

$$Q = \frac{2\pi k L (T_1 - T_o)}{\log_e \frac{r_2}{r_1}} = \frac{T_1 - T_o}{\frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k L}} = \frac{T_1 - T_o}{R}$$

Where, $R = \frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k L}$ conductive Resistance

Heat transfer through composite cylinder with conduction and convection :

$$Q = h_i 2\pi r_1 L (T_1 - T_i) = \frac{2\pi k_1 L (T_1 - T_2)}{\ln\left(\frac{r_2}{r_1}\right)} = \frac{2\pi k_2 L (T_2 - T_3)}{\ln\left(\frac{r_3}{r_2}\right)}$$

$$= h_o 2\pi r_3 \cdot L (T_3 - T_o)$$

$$Q = \frac{T_1 - T_o}{R_1 + R_2 + R_3 + R_4} = \frac{(T_1 - T_o)}{\frac{1}{h_i 2\pi r_1 L} + \frac{\ln\left(\frac{r_2}{r_1}\right)}{2\pi k_1 L} + \frac{\ln\left(\frac{r_3}{r_2}\right)}{2\pi k_2 L} + \frac{1}{h_o 2\pi r_3 L}}$$

Heat transfer through a hollow sphere :

$$Q = \frac{4\pi k (T_1 - T_o)}{\left(\frac{1}{r_1} - \frac{1}{r_2}\right)} = \frac{(T_1 - T_o)}{\left[\frac{(r_2 - r_1)}{4\pi k r_1 r_2}\right]} = \frac{T_1 - T_o}{R}$$

- Heat transfer through a compound sphere with convection

$$Q = \frac{T_1 - T_o}{\frac{1}{A_1 h_1} + \frac{(T_1 - T_2)}{4\pi k_1 r_1 r_2} + \frac{(T_2 - T_3)}{4\pi k_2 r_2 r_3} + \frac{(T_3 - T_o)}{A_o h_o}} = \frac{(T_1 - T_o)}{\frac{1}{A_o h_o}}$$

Where, $A_1 = 4\pi r_1^2$, $A_o = 4\pi r_3^2$

$$R_1 = \frac{1}{A_1 h_1}, \quad R_2 = \frac{(r_2 - r_1)}{4\pi k_1 r_1 r_2}, \quad R_3 = \frac{(r_3 - r_2)}{4\pi k_2 r_2 r_3},$$

$$R_o = \frac{1}{A_o h_o}$$

$$Q = \frac{T_1 - T_o}{R_1 + R_2 + R_3 + R_o}$$

Section II : Insulation

- A material having low thermal conductivity is called **insulating material**. It is used to decrease the heat transfer rates.

Examples of insulating materials are asbestos, glass, rock wood, cork, plastic materials etc.

- Critical thickness of insulating material**

$$(a) \text{ For cylinders : } r_c = r_o = \frac{k}{h}$$

$$(b) \text{ For spheres : } r_c = r_o = \frac{2k}{h}$$

- For reducing heat loss in cylinder due to insulation

$$R_{\text{insulation}} \gg r_o$$

Exercise

Q.1 Discuss "Electrical analogy" of combined heat conduction and convection in a composite plane wall. Hence define overall heat transfer coefficient.

(Section 2.2 and 2.3)

Q.2 Explain the analogy between heat conduction and electricity. (Section 2.2)

Q.3 What do you understand by thermal contact resistance? How does it affect the rate of heat transfer. (Section 2.6)

Q.4 Derive an expression through a composite cylinder with conduction and convection. (Section 2.7)

Q.5 Derive an expression through composite sphere with conduction and convection. (Section 2.8)

Q.6 Explain different types of insulating materials and their desirable properties. (Section 2.9)

Q.7 Explain critical radius of insulation. (Section 2.10)

Q.8 Derive an expression for critical radius of insulation for a cylinder. (Section 2.11)

Q.9 Derive an expression for critical radius of insulation for a sphere. (Section 2.12)